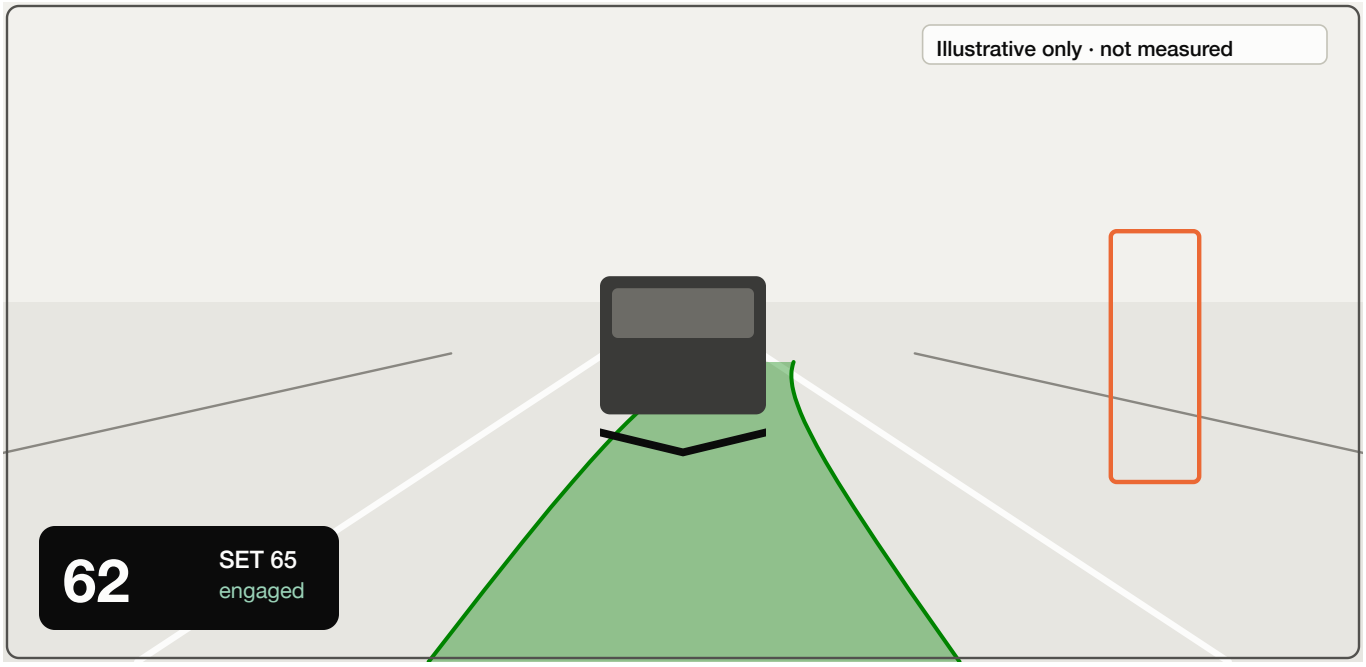
Windshield AR HUD

A road-registered augmented-reality head-up display built on openpilot, with openpilot providing adaptive cruise and lane centering in the same car.

This revision assumes the budget is not the constraint: every purchase carries a one-line justification, and the design aims at the best demo the physics allows.

Revision 2 · PDF edition · written 2026-10-08, revised 2026-10-09.
Research sections were drafted by Claude Sonnet 5.5 agents and reviewed, corrected and assembled by Claude Fable 5.1. This PDF edition expands the plan, adds explanatory figures and corrects several numbers (see the next page); it was prepared by Claude Haiku 5.5.
Facts that could not be verified today are marked **UNVERIFIED**. Anything learned about the Jetson is from 2026-10-07; the board was unreachable when the research was written.



Illustrative driver's-eye view of the HUD window: drive path, lane lines, lead chevron, speed block and a hazard outline. Not a measured image.

About this edition

This is the same plan as revision 2, rewritten as a document to read on paper or on screen. Every section now explains the mechanism behind a decision before it gives the number. Where a mechanism was previously stated only as a result, it is now derived in Appendix A, so a reader can check the arithmetic.

Figures are schematics unless their caption says otherwise. Those that plot numbers are computed from the same values as the text, so a changed input changes both. Heights and distances in the schematics are labelled as true or not to scale.

Changes from revision 2

Checking the plan's arithmetic against its own assumptions turned up the following. The text, figures and appendices in this edition use the corrected values. Anyone who worked from revision 2 should read these first.

1. **Relay mirror geometry.** Revision 2 put the PGU at 346 mm from a mirror with f ≈ 330 mm. Since 346 mm is outside the focal length, that single-mirror equation gives a real image at 7.1 m, not a virtual image. A virtual image at 7.5 m needs the PGU inside the focal length, at 316 mm. That gives magnification 23.7 and a 10.8° × 5.4° field. The windshield is still not in this model, so the mirror is not ready to order (see Stage 2 in the optics chapter).
2. **Dimming range.** Revision 2 called the night requirement a "300:1 dimming range". From the 1,500 cd/m² noon target down to the 50 cd/m² night ceiling the ratio is 30:1. If the projector runs at its full output, 42,441 cd/m² at the eye, the range down to 50 cd/m² is 849:1.
3. **Luminance table.** The "after 70/30 combiner" and "after windshield" columns are two different light paths, not a cascade. The windshield column was computed from the source value times 24%. This edition labels the paths separately and adds a row with an assumed 90% reflectance for each of the two relay mirrors.
4. **Demo window shape.** Section 2 of revision 2 gave the window as 10° × 4°, but the PGU is 2:1 (60 × 30 mm), and the optics section sized the image for 10° × 5°. A 10° × 4° window needs a 2.5:1 active region, which means cropping the PGU. This edition states the window as 10.8° × 5.4° (matching the corrected mirror) with 10° × 5° as the alternative.
5. **Prediction claim.** Revision 2 said the residual error at 100 ms is "a fraction of a degree". Both halves are now shown: without prediction the 3 m of travel at 30 m/s moves a road point 10 m ahead by 2.9°; with prediction the residual is 0.003° for 1 m/s² of unmodelled acceleration.
6. **Parallax table.** A 7.5 m row was added, since that is where the virtual image sits: 1.15° for a 150 mm head shift.
7. **Citations.** The SPI handle in `panda.cc` is at line 16 (revision 2 said 15). The modeld device branch in `SConscript` is at lines 27–31 (revision 2 said 28–32).

The latency total is the rounded form of the stage ranges, which sum to 67–139 ms; the plan's "70–140 ms" is kept where it is quoted as the plan's figure.

Revised 2026-10-09

A second check of the PDF edition against the physics and the source tree. The repository line references all held; the changes are in the optics and the hardware notes.

1. **Polarisation in the luminance budget.** The rear-projection diffuser depolarises the projector's light, so the polariser sheet in the parts list is needed and passes about 42% of it. With that loss the daylight headroom through the windshield falls from 23× to 10× (section 6.5, Appendix A.5). The LCD monitor is already polarised, so its mounting orientation decides whether the windshield reflects 23.6% or 1.2% of its light (section 6.3).
2. **Polarised sunglasses.** An s-polarised image is largely blocked by polarised sunglasses. Added as a risk with the demo-day mitigation (section 11.1).
3. **Sun focus in watts.** The concentration ratio is now accompanied by the collected power, about 42 W into a spot a few millimetres across (section 6.7, Appendix A.12).
4. **Dimming floor.** The daylight working point is about 10% of full projector output on the realistic path. Whether the projector can dim that far is unverified, so a fixed neutral-density filter is now named as the fallback (section 6.5).
5. **Two-surface reflection.** The 24% windshield factor is for one surface; with the wedge overlapping both images it is about 37%, so the budget is conservative (Appendix A.5).
6. **Head tracker wording.** MediaPipe Face Landmarker runs on the Jetson, not on the OAK-D Pro; running a converted model on the camera's own processor is an option to verify (sections 7.1, 8.2).
7. **Citations.** The `process_config.py` lines for Track B were checked: line 83 is time synchronisation, not thermal management, and `hardwared` does run on PC. The blocker is now stated as IMU and GPS only (section 5.3, Appendix C).

How to read the tags and units

MARKER	MEANING
UNVERIFIED	A fact, price or product claim that could not be checked when the plan was written. Do not rely on it without checking.
VERIFY	A condition that depends on hardware or software behaviour not yet tested on this device, such as whether a USB port runs in host mode.
CONFIRM LOCALLY	A legal point that depends on the jurisdiction. The plan gives the current position, not legal advice.
FIX	A value or claim corrected in this edition. Each one is listed above.
ADDED	A risk or open question added in this edition, found while checking the plan. Revision 2 does not contain it.

Units: luminance in cd/m² (candela per square metre, also called a nit); angles in degrees unless stated as mrad; distances in metres or millimetres as written; frequencies in Hz; latency in milliseconds. Monetary values are US dollars as listed in the plan.

CONTENTS

	About this edition	2
1	Recommendation in brief	4
2	Background for readers new to the project	4
3	What the demo shows	10
4	System architecture	12
5	Self-driving integration	19
6	Optics	21
7	Software	32
8	Hardware roles and acquisitions	35
9	Phases and acceptance criteria	40
10	Validation and demo script	44
11	Risks, legal, open questions	47
A	Derivations	50
B	Glossary	54
C	Repository pointers	56
D	Sources	57

1 Recommendation in brief

Split the system into two computers. A comma four runs openpilot and drives the car. The Jetson Orin NX runs only the HUD. It subscribes to openpilot's outputs over a wired link, tracks the driver's head, re-projects the model's path, lane lines and lead vehicle into the driver's eye, and draws them on a picture generation unit (PGU). A mirror relays that picture into the windshield.

The HUD never sits in the control path. The only device that can put bytes on the car's CAN bus is the panda inside the comma four, and its firmware limits torque and acceleration on every frame whatever the Jetson does. Power-cycling the Jetson mid-drive must leave openpilot unaffected; that is one of the acceptance tests in section 10.

1,500	10.8° × 5.4°	67–139 ms
cd/m² noon target at the eye, for a contrast ratio of 1.15 against 10,000 cd/m² sky	demo window for the corrected relay design, with the virtual image at 7.5 m	uncompensated latency across the four timed stages, before the HUD predicts forward

Why not run openpilot on the Jetson?

Four things in this checkout make that a multi-month port with no HUD benefit:

1. The driving model is compiled only for Qualcomm, Metal or CPU. There is no CUDA path.
2. The C++ panda daemon speaks SPI only, so an external USB panda does not work without new code.
3. The PC build has no IMU or GPS processes.
4. No purchasable GMSL camera matches comma's image sensors with a JetPack 6 driver.

The comma four removes all four problems for $999 (the plan's price, read 2026-10-08). The Jetson's GPU is better spent on the HUD's own perception and rendering. Running openpilot on the Jetson stays in the plan as a research track, Track B, with its blockers listed in section 5 rather than as the critical path.

Optics are the hard part

A flat display reflected in a windshield forms its image about a metre away, so road-registered graphics cannot be drawn on it without the driver refocusing. Production AR head-up displays place the virtual image 7.5 to 10 m out with a concave mirror. Daylight readability needs roughly 1,500 cd/m² reaching the eye.

A 1,200-nit monitor delivers about 19% of that target after the windshield, so monitors are used for the bench rig and night driving. A 2,000-lumen laser DLP projector, shining onto a small diffuser, delivers about 28 times the daylight target through the windshield path, about 23 times with two relay mirrors at an assumed 90% reflectance each, and about 10 times once the polariser sheet that the diffuser makes necessary is included. It is the daylight picture generator in the final stage. Section 6 gives the full budget.

Three decisions needed before ordering

Which car: it must be on openpilot's supported list, and it should have an OEM head-up display trim so a wedge-PVB windshield exists for it. The car also fixes the windshield's curvature, which the relay mirror's design depends on, so the mirror cannot be specified until the car is chosen. Where the demo will be driven: a closed course or private lot. Who signs off the vehicle modification at the university.

Everything else in this plan can start on the bench today.

How the plan is organised

CHAPTER	WHAT YOU LEARN FROM IT
2 Background	What a head-up display is, what a virtual image is, what registration means, and the openpilot pieces the HUD reads.
3 Demo	The six things the driver sees, why each is there, and what is still undecided.
4 Architecture	The two computers, the safety boundary, the message transport, latency, prediction, and head tracking.
5 Self-driving	Track A (the critical path) and Track B (openpilot on the Jetson) with their go/no-go gates.
6 Optics	The constraints, the reflection physics, the luminance budget, three build stages, the sun hazard and calibration.
7 Software	The new processes, the projection chain, age compensation and bring-up.
8 Hardware	What is owned, what to buy and why, wiring and power, and the order of purchases.
9 Phases	The twelve-week critical path, phase by phase, with acceptance criteria.
10 Validation	Measurable targets, how each one is measured, and the five-minute demo script.
11 Risks	Risks with mitigations, the legal position to confirm, and open questions.
A–D	Derivations of every number, a glossary, repository pointers and sources.

2 Background for readers new to the project

This chapter explains the four ideas the rest of the plan relies on: what a head-up display does, where its image appears and why that matters, what registration means, and what openpilot exposes to other programs. A reader who already knows these can skip to chapter 3.

2.1 What a head-up display does

A head-up display (HUD) puts information in the driver's line of sight so the driver does not have to look down at a dashboard. There are two design questions. First, where the image appears to be, meaning its focus distance. Second, how large it is, meaning its field of view, the angle it covers as seen from the eye.

An augmented-reality HUD goes further: it places each graphic where the thing it describes is. A path drawn on the road, or a box drawn around a pedestrian, only works if the overlay stays on the object as the car moves and the driver's head moves. The technical word for that is *registration*, covered in section 2.3.

2.2 Where the image appears

Light from a display reaches the eye as a bundle of rays. If the rays diverge as if they came from a point 7.5 m away, the eye perceives the image at 7.5 m. Such an image is called a *virtual image*: no light actually originates there, but the eye cannot tell. A virtual image is the mechanism that lets a HUD draw content at road distance.

The eye focuses on whatever it is looking at, a process called accommodation. A screen 0.6 m away forces the eye to refocus when the driver looks up at the road, which can take a fraction of a second and causes eye strain. A virtual image at 7.5 m is in the range where road objects sit, so the eye can keep one focus across the road and the graphics.

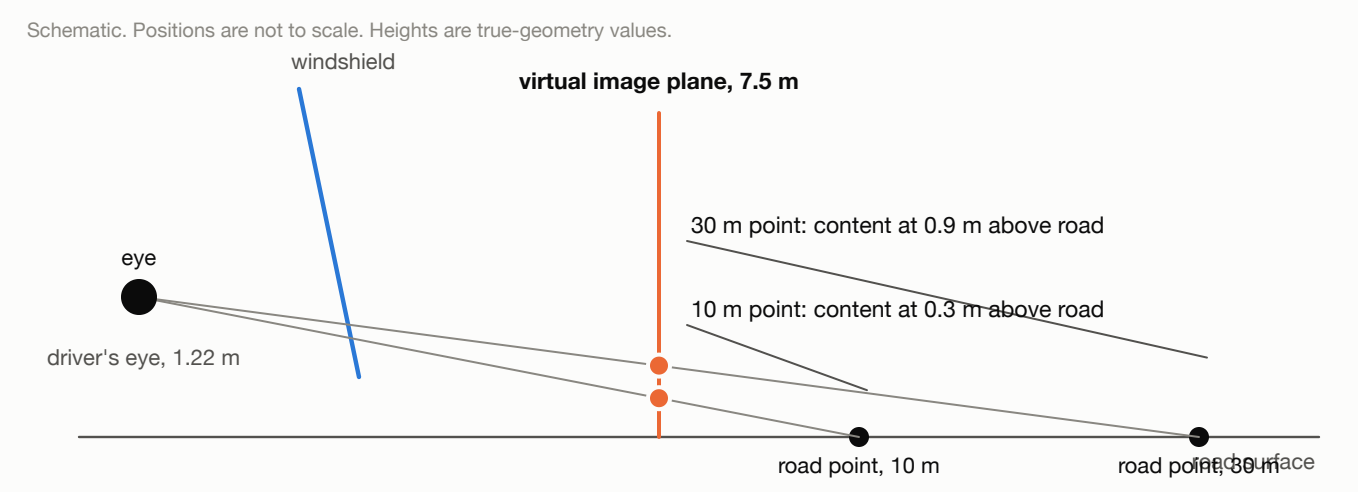


Figure 1. Schematic of a virtual image plane between the eye and the road. To draw a mark on the road point, the HUD places it where the line from the eye to the point crosses the plane. Positions are not to scale; heights in the captions are true-geometry values.

The plan's convention is that a road point at distance d is drawn where the line of sight crosses a plane at distance D. On a flat road with the eye at height h = 1.22 m, the crossing is at height $h(1 - D/d)$ above the road. With D = 7.5 m, a point 10 m away is drawn 0.31 m above the road on the plane, and a point 30 m away is drawn 0.92 m above it. Far points therefore sit higher on the plane, and their marks shift less when the head moves.

2.3 Registration and conformal graphics

Registration error is the angle between where a graphic is drawn and where the thing it marks actually is, measured from the eye. A one-degree error at 30 m is about half a metre on the road. The plan calls graphics that must register *conformal*, and graphics that stay fixed *non-conformal*. Each element in the demo is one or the other.

ELEMENT	WHERE IT APPEARS	MUST IT REGISTER?
Drive path	On the road surface, along the lane	Yes. Its error is visible against the lane paint.
Lane lines and road edges	Over the real paint	Yes. Drawn only where the model is confident.
Lead vehicle chevron	Under the lead car's rear bumper	Yes. Its gap is read by the driver as the car ahead's position.
Hazard outlines	Around a detected pedestrian, sign or cyclist	Yes. HUD-only; never feeds control.
Speed, set speed, engagement	Small fixed block low in the window	No. This is the only non-conformal content.
Driver-monitoring warnings	Not in the AR window	No. They stay on the comma four's own screen.

Registration is the reason the rest of the plan is so detailed. A path that is 1° off at 30 m looks wrong to a driver who is watching the lane paint, even though the same error would be invisible on a dashboard gauge.

2.4 openpilot in brief

openpilot is an open-source driver-assistance system for cars on its supported list. It provides adaptive cruise control (ACC), which holds a set speed and a gap to the car ahead, and lane centering, which steers within the lane. The comma four is the comma device that runs it; the plan uses it unmodified for driving.

openpilot is a set of separate programs that talk through a message bus. Cameras feed a driving model, the model's output feeds a planner, the planner feeds a controller, and the controller's commands go through an interface to the car. Each program publishes typed messages on named services, and any program that subscribes receives the latest message. The HUD uses the same mechanism: it is one more subscriber, and it never publishes a control message.

The services the HUD reads, with their rates from `cereal/services.py` and a plain-language summary of each field group, are below. The full field definitions are in `cereal/log.capnp`.

SERVICE	RATE	WHAT IT CARRIES
`modelV2`	20 Hz	The driving model's output: predicted path, lane lines and road edges, lead predictions. Timestamped at the camera frame (`timestampEof`).
`radarState`	20 Hz	Lead vehicles from radar, where the car provides radar.
`longitudinalPlan`	20 Hz	Planned speed and acceleration from the longitudinal planner.
`carState`	100 Hz	Vehicle state from the car bus: speed, yaw rate, steering and related signals.
`controlsState`	100 Hz	State of the lateral and longitudinal controllers.
`selfdriveState`	100 Hz	Whether openpilot is engaged and in which mode. Drives the path's colour.
`extrinsicsCalibration`	4 Hz	The camera's calibrated position in the car frame.
`pandaStates`	10 Hz	State of the panda and the car interface.
`deviceState`	2 Hz	Device health and temperature.

2.5 The safety model and supported cars

The panda is a small board between the computer and the car's CAN bus, the wiring over which the car's electronic control units talk. The panda's firmware checks each message the computer asks to send against a safety model written for that car. A command outside the model's limits is blocked before it reaches the bus. This is why the plan treats the panda as the safety boundary: it is the last check, in hardware, that does not depend on the software that asks for control.

A *supported car* is one for which openpilot has a tested interface and safety model. The plan selects the car in Phase 0 from the repository's car list in `docs/CARS.md`, preferring a car with a factory HUD trim and openpilot longitudinal control.

3 What the demo shows

Everything else in the plan is derived from this list. In a window above the hood line, registered to the real road, the driver sees six things. The window is 10.8° × 5.4° in the corrected design; revision 2 gave 10° × 4°, and the optics chapter explains why the shape changed.

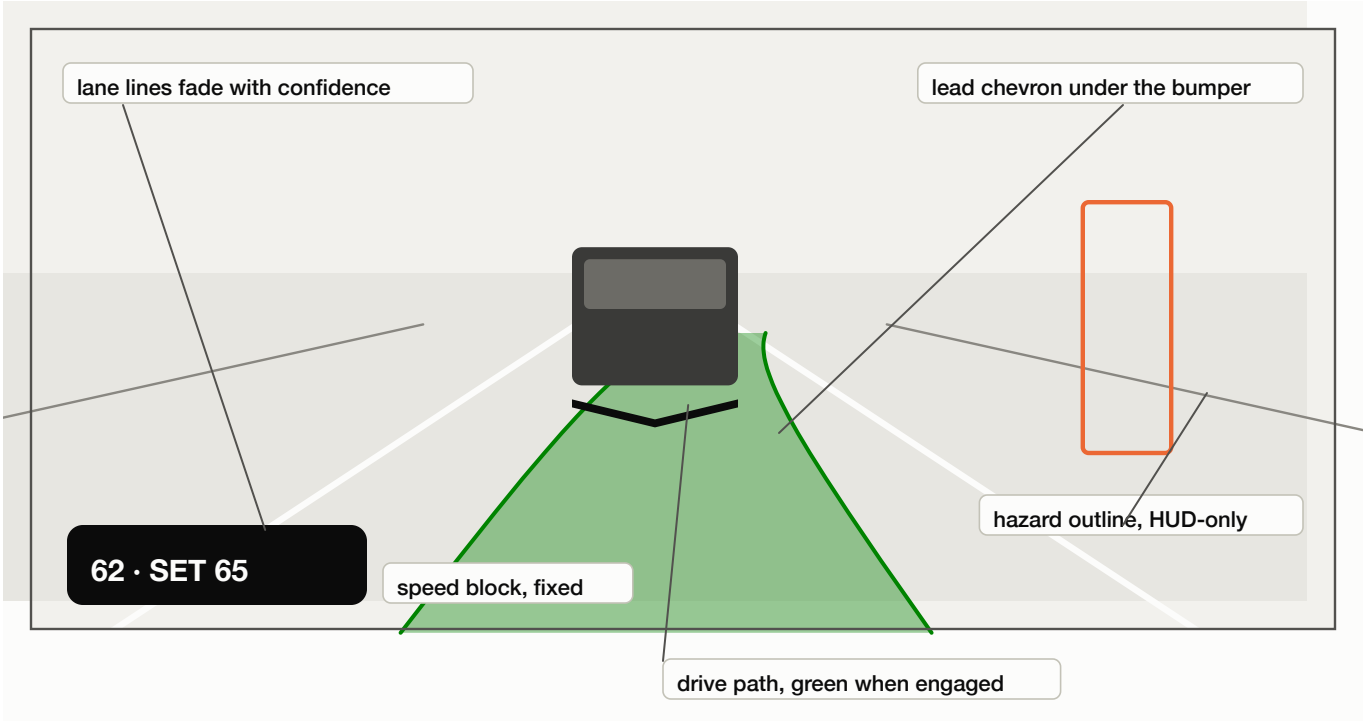


Figure 2. Mockup of the demo window with the five registered and fixed elements labelled. The hazard outline is HUD-only. Proportions follow a flat-road perspective at a 1.22 m eye height; the content is illustrative, not a measured image.

3.1 The six items

1. **Drive path.** openpilot's planned path as a translucent ribbon lying on the road surface, tinted by engagement state. The demo script says the ribbon turns green when openpilot engages. This is the most convincing element, because any error is visible against the real lane paint.
2. **Lane lines and road edges.** Thin lines over the real paint. They are drawn only when the model is confident, and they fade with the lane-line probabilities instead of popping on and off.
3. **Lead vehicle marker.** A chevron under the lead car's rear bumper, with the following gap in metres and seconds. Studies of attention in driving suggest the lead vehicle is where attention should stay, so cues go there rather than across the whole screen.
4. **Speed, set speed and engagement.** A small, fixed block low in the window. It is the only non-conformal content, so it is the only element that does not move with the scene.
5. **Hazard cues from the Jetson's own perception.** Pedestrians and signs detected by a TensorRT network on the Jetson, highlighted as outlines. These are HUD-only. They never feed control, and openpilot does not know they exist.

6. **Head tracking made visible.** The passenger-side monitor shows the rendered frame and the driver camera with the eye point overlaid, so the audience can watch the parallax correction work as the driver leans.

3.2 Where each item's data comes from

ITEM	INPUT	STATUS	BUILT BY
Drive path	`modelV2` path, `selfdriveState`	Exists	`hudd` renderer, new
Lane lines	`modelV2` lane lines and probabilities	Exists	`hudd`, new
Lead chevron	Lead predictions in `modelV2`; `radarState`	Exists	`hudd`, new
Speed block	`carState`, `selfdriveState`	Exists in the UI	Reuses `hud_renderer.py` fields
Hazard outlines	Detector on the Jetson camera	New	`perceptd`, publishes `hudDetections`
Head position	OAK-D Pro, MediaPipe Face Landmarker	New	`headtrackd`, publishes `hudEyePose`

The two new services, `hudDetections` and `hudEyePose`, are not in `cereal/services.py` yet. Adding them is a Phase 1 task.

3.3 What is deliberately left out

- **Navigation arrows.** openpilot's navigation messages are deprecated in this checkout, so there is no reliable source.
- **Driver-monitoring warnings in the AR window.** They stay on the comma four's screen, where the existing UI already shows them.
- **Full-windshield clutter.** Anything not conformal or safety-relevant is removed or dimmed. The clutter mode in the demo is a deliberate contrast, shown for ten seconds.

> **What the plan does not yet specify**
>
> These behaviours matter for safety and must be decided and tested in Phase 1, before any car drive. The plan does not currently define them.
>
> - What the HUD shows when a source message stops arriving, whether because of the link, the bridge or a crashed process. The plan does not set a stale-data cutoff.
> - Whether the drive path fades out or disappears when openpilot disengages, and what the speed block shows during the handover.
> - The probability thresholds at which lane lines begin and finish fading.
> - The minimum confidence and persistence for a hazard outline, and how a hazard that leaves the view is removed.
> - How the demo's clutter mode is defined: which elements appear, and at what brightness.

4 System architecture

4.1 Two computers, one direction

The architecture has two computers and one data path. The comma four is the driving computer. It is the only computer connected to the car's bus, and it runs openpilot unchanged. The Jetson is the HUD computer. It receives a copy of openpilot's outputs over Ethernet and renders the HUD. Nothing flows back. The optical system at the end is driven only by the Jetson's HDMI output.

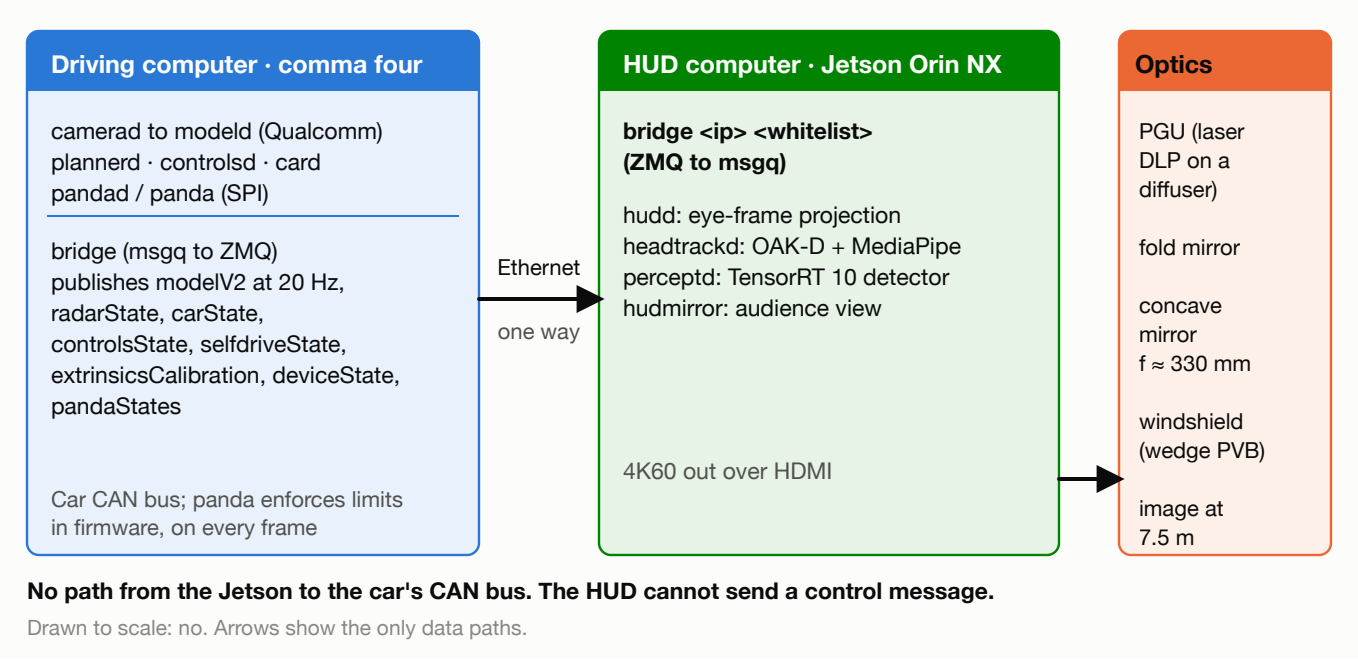


Figure 3. Two-computer architecture. Data flows one way from the driving computer to the HUD computer, then to the optics. The arrows show the only data paths.

The split has four reasons, each tied to a fact in the checkout:

- **The safety boundary is physical.** The panda in the comma four enforces the opendbc safety model on every transmitted frame. The Jetson has no CAN connection at all, so no software fault on the Jetson can reach the bus.
- **Calibrated sensing comes with the comma four.** Its cameras, IMU and GPS are what `DEVICE_CAMERAS`, calibrationd and locationd expect. The webcam path in this tree captures 1280 × 720, a size for which no model warp file exists, and it reports no sensor name, so it silently assumes the comma camera's 2648 px focal length. Using a webcam on the Jetson would mean building a new calibration path.
- **The transport already exists.** `openpilot/cereal/messaging/bridge` republishes every service over ZMQ on a per-service port. The same binary, run with an IP address and a whitelist, subscribes and republishes locally as msgq. A HUD process on the Jetson then uses a plain `SubMaster` with no changes to the messaging code.
- **The Jetson's GPU has a real job.** 4K60 rendering with pre-distortion, head tracking at camera rate, and a detector on TensorRT 10 all fit the Orin NX. None of them fit the comma four's spare capacity.

4.2 The safety boundary

The safety argument has to be easy to state, so the plan separates two paths that meet only at a one-way link. The control path is the chain that decides what the car does. It ends at the panda, which checks every frame. The HUD path reads a copy of messages and draws pixels. Nothing in the HUD path can send a command, and the Jetson has no electrical path to the bus.

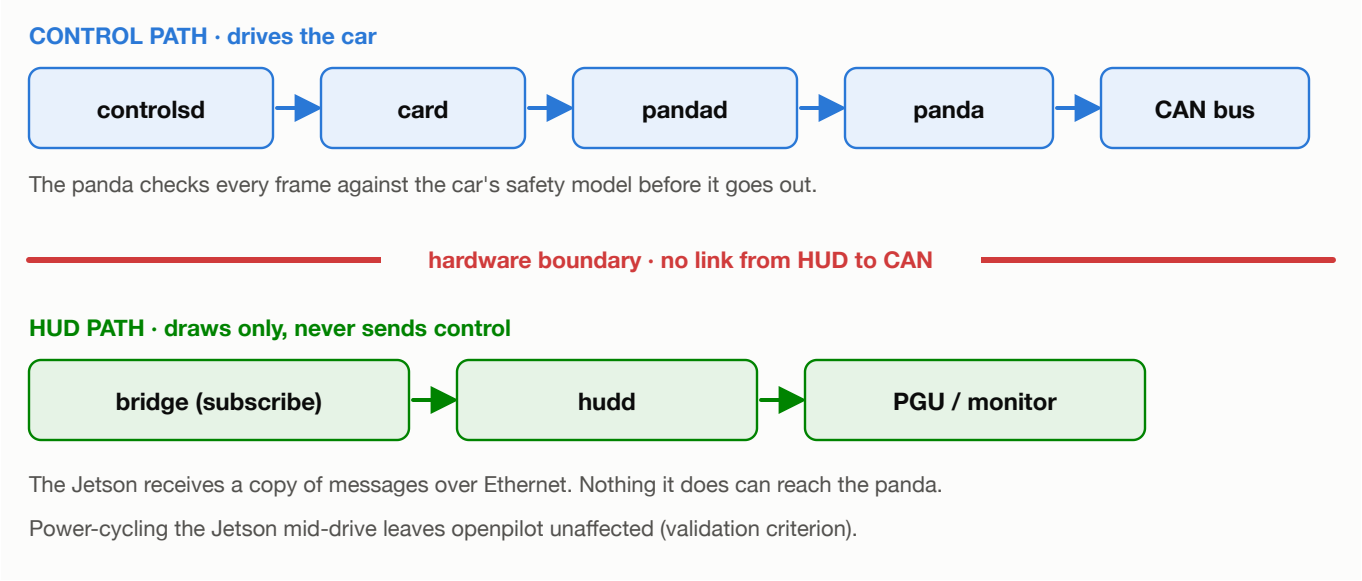


Figure 4. The control path and the HUD path. The only link between them is the bridge's copy of messages, which flows one way. The panda is the final check on every control frame.

Three details follow from this design. The bridge on the comma four publishes a copy of messages that openpilot already publishes; it does not write to the panda. Today `process_config.py` starts the bridge only for `notCar` builds, so the plan adds a Params flag that enables it in a car (section 5). Second, if the Jetson stops responding, the car is unaffected, because the Jetson was never in the loop. Third, the acceptance test for this boundary is physical: a power cycle of the Jetson on a closed course, with the openpilot drive continuing unchanged (section 10).

4.3 How messages travel

On the comma four, messages move between programs as msgq, a local inter-process queue. Across a network, msgq cannot be used directly. The bridge converts between the two. On the sending side it subscribes to local msgq services and publishes them over ZMQ, a network messaging library. Each service gets its own network port, derived from a hash of the service name. On the receiving side a second bridge instance subscribes to those ZMQ ports and republishes locally as msgq, so the HUD program on the Jetson sees ordinary local messages.

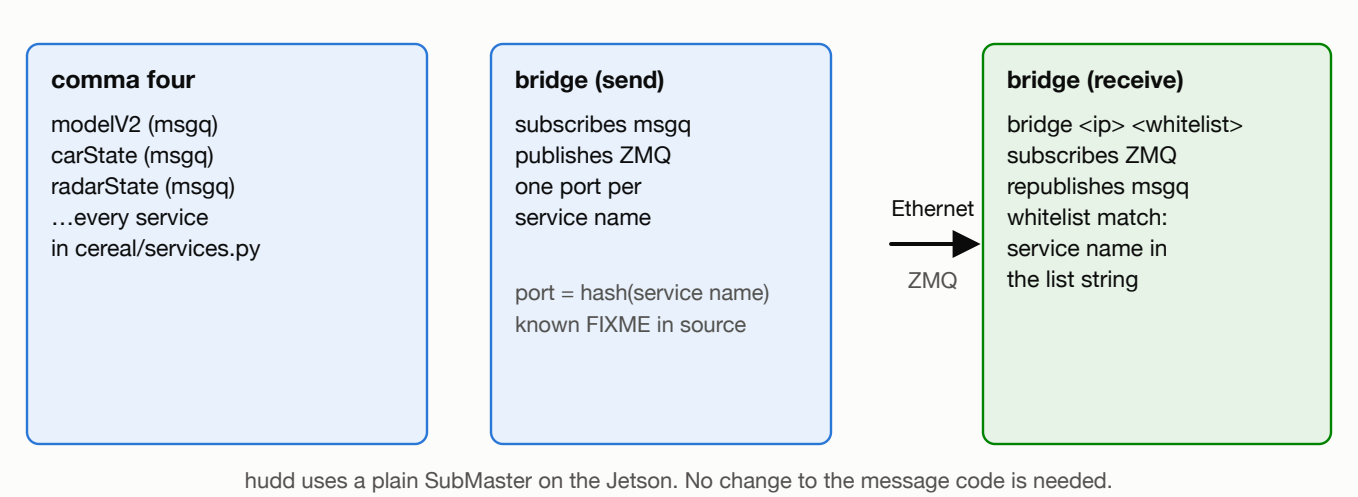


Figure 5. Message path from the comma four's local queues, across Ethernet as ZMQ, back into local queues on the Jetson.

- **Ports.** The port for each service is hashed from its name, and the hashing has a known FIXME in the source. Collisions are unlikely with a short list of services, so the plan checks them once at setup rather than designing around them.
- **Whitelist.** On the receiving side the bridge is started with a whitelist string. In `bridge.cc` line 13 a service is accepted when its name appears anywhere inside the whitelist string. A short name that sits inside a longer one is therefore accepted too, so the whitelist should use exact, full service names.
- **Stale code.** The `ZMQ=1` environment flag and `SubMaster(addr=...)` are dead in this checkout: msgq asserts that the address is 127.0.0.1. `tools/camerastream` and `replay/ui.py` depend on them. Video over the link is not needed for the HUD, so the plan notes this rather than fixing it.

4.4 Message timing

Each service has its own rate, set when it is published. The HUD does not need to match them. A `SubMaster` keeps the latest message from each service and reports how old it is. hudd reads the newest `modelV2` for the path, the newest `carState` for speed, and so on, and it renders at 60 Hz regardless. A model message at 20 Hz is therefore drawn three times at the 60 Hz render rate, and each draw is advanced by that message's measured age (section 4.6).

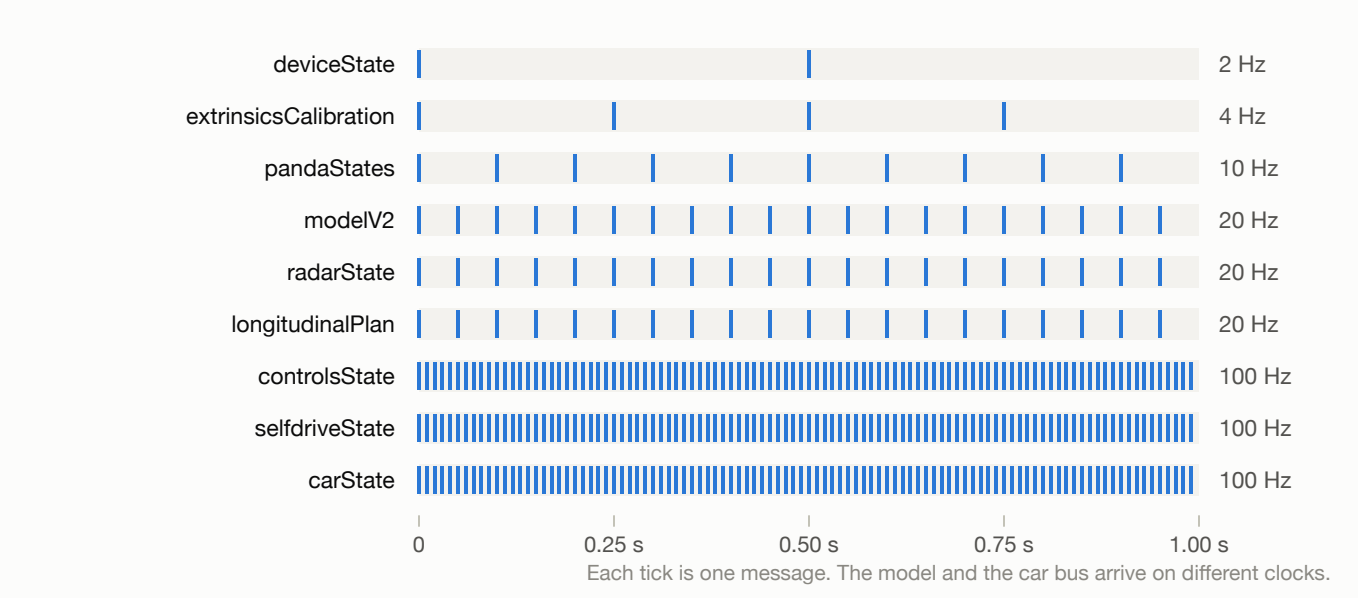
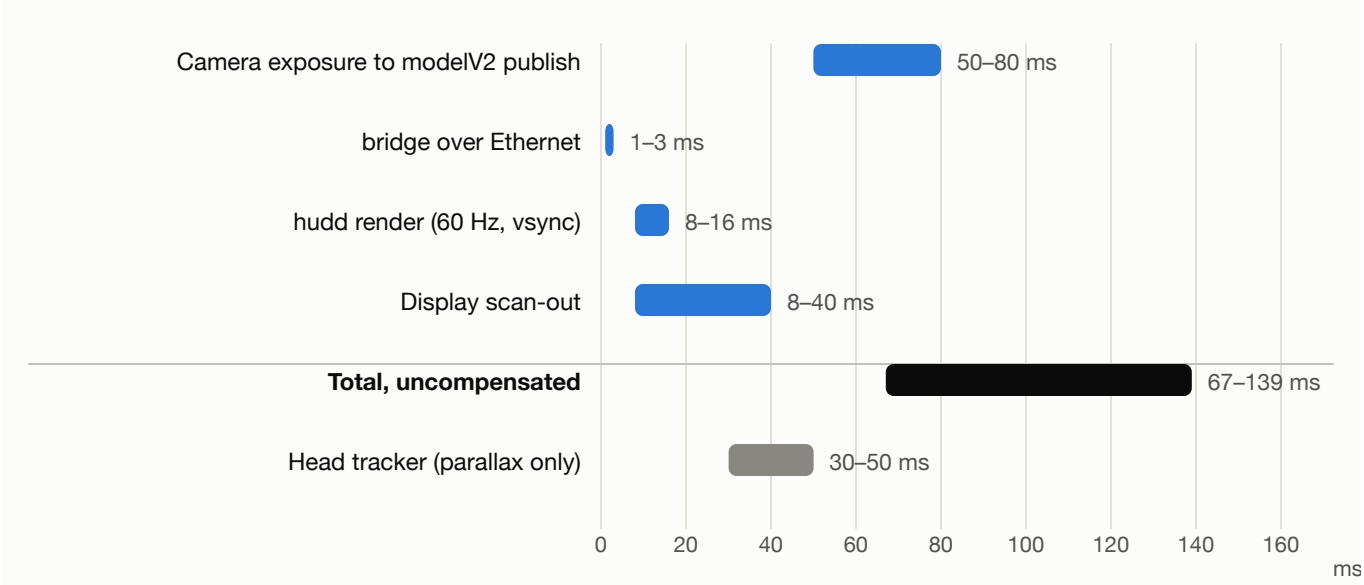


Figure 6. Publish rates of the services the HUD reads, one second of messages. Each tick is one message. Rates are from `cereal/services.py`.

The timing picture explains why the age of the model's output is the key quantity. The path comes from a camera frame that is already old when the model publishes it, while the car's speed arrives at 100 Hz and is newer. The HUD combines them, so the path and the speed may describe different moments. Measuring the path's age from the camera frame, and not from when the message arrived, keeps the two consistent.

4.5 Latency budget

Road-registered content is convincing only if it stays put when the car moves. The plan's target is set by the difference between when a picture is drawn and when the light reaches the eye, a delay called motion-to-photon latency. The literature puts the perceptual threshold for head-mounted AR near 20 ms, and university-project targets for automotive head-up displays near 50 ms. The stages below are the ones the plan measures; the head tracker is shown separately because it affects parallax, not the road-frame content.



The plan rounds the total to 70–140 ms. The exact sums of the stage ranges are shown.

Figure 7. Typical age of each stage, as ranges. The total is the sum of the four timed stages, from the camera exposure to the display; the plan rounds it to 70–140 ms.

STAGE	TYPICAL AGE	NOTES FROM THE PLAN
Camera exposure to modelV2 publish	50–80 ms	20 Hz model, inference on the comma four. Measured via `modelV2.timestampEof`.
bridge over Ethernet	1–3 ms	Wired. Wi-Fi adds jitter, so avoid it.
hudd render	8–16 ms	60 Hz. Vsync on to avoid tearing in the mirror.
Display scan-out	8–40 ms	The monitor is fast. Projector input lag must be measured and is the likely worst stage.
Head tracker	30–50 ms	Affects parallax only, not the road-frame content.
Total, four stages, uncompensated	**67–139 ms**	At 30 m/s this is 2–4 m of forward travel.

The exact sums in the table are 67 to 139 ms. The plan's 70–140 ms is the rounded form. At 30 m/s, the 67 to 139 ms span is 2 to 4.2 m of travel, which is the distance by which an uncompensated image would lag the road.

4.6 Predicting forward

The image cannot be made faster than the pipeline, so the HUD predicts instead. Each point on the model's path is in the car's frame, with x forward, y to the right and z down. The car has moved since the camera frame was taken, so before projection hudd moves each point to where it is now in the car's current frame: back by the distance the car has travelled, `vEgo` × age, and rotated by the yaw the car has turned through, `yawRate` × age. Age is the current time minus `modelV2.timestampEof`, after the clocks of the two computers are aligned over NTP on the private link.

```
age = t_now − modelV2.timestampEof
p_now = Rz(−yawRate · age) · (p_model − [vEgo · age, 0, 0])
```

The plan's wording says the path is "advanced" by `vEgo · age`. The sign above follows from the car moving forward, so the path points move toward the car. Confirm the sign and the rotation direction on a replayed route before relying on either.

The prediction matters a great deal, and the size of its effect is easy to misjudge, so the figure shows both cases. Take a road point 10 m ahead, a camera 1.22 m above the road, and a car at 30 m/s. In 100 ms the car moves 3 m. Without prediction, the point's angle below the horizontal changes from 6.96° to 9.89°, which is a shift of 2.9° as seen by the driver. That is an obvious error. With prediction, the remaining error comes only from changes the model did not foresee. Given 1 m/s² of unmodelled acceleration over the same 100 ms, the error is 5 mm, which is 0.003° at 10 m. The plan's target is 0.5° static and 0.3° under head motion (section 10).

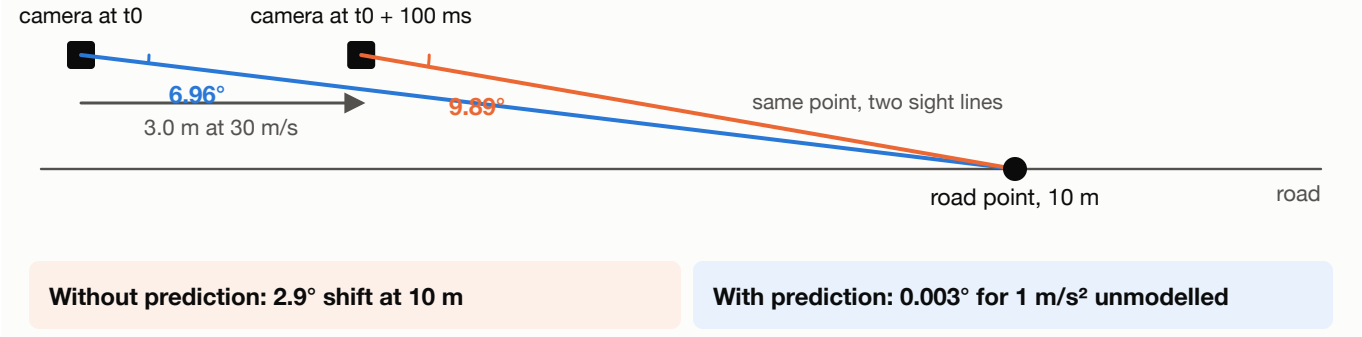


Figure 8. Side view drawn to true scale. The camera moves 3 m in the time the image is delayed, so the same road point is seen at a steeper angle. The residual figures are for the stated unmodelled acceleration.

Prediction has limits, and the plan states them. It corrects for the car's own motion along the path. It does not correct for another road user's motion, for a model that changes its mind, or for a sudden change in acceleration. `MIN_DRAW_DISTANCE` rises from 10 m to the bottom edge of the virtual window, so nothing is drawn where the hood hides the road.

4.7 Head tracking and parallax

The virtual image is fixed in space, but the driver's eye moves. A graphic computed for the nominal eye position therefore lands in a different place on the road when the eye moves. The shift is called parallax. The head tracker measures where the eye is so hudd can compute the mark's position from the actual eye.

Parallax is largest for near objects. A 150 mm head shift, which is the plan's test span, moves a point 10 m away by 0.86° as seen from the eye, and a point 30 m away by 0.29°. The head tracker removes most of that shift when it works.

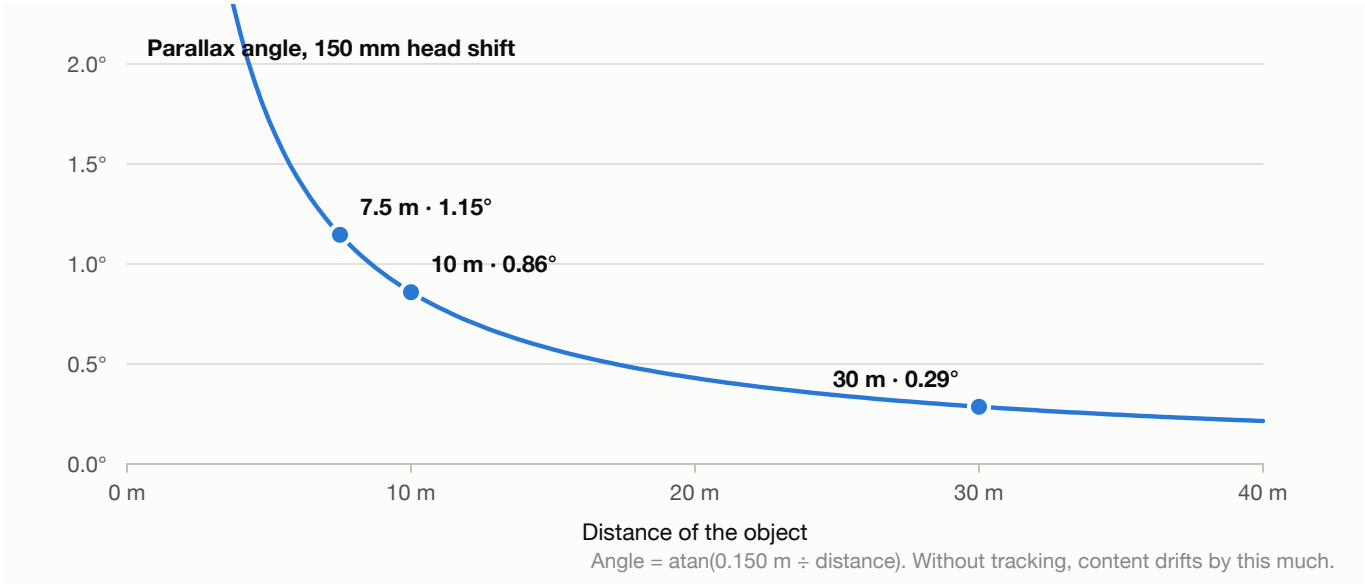


Figure 9. Parallax angle against object distance for a 150 mm head shift, with the plan's distances marked. The curve falls quickly with distance, so near content needs the tracker most.

OBJECT DISTANCE	PARALLAX FOR 150 MM HEAD SHIFT	WHERE IT MATTERS IN THE PLAN
7.5 m	1.15°	The virtual image's own distance. Its content sits on this plane.
10 m	0.86°	Nearest path points and the lead chevron in traffic.
15 m	0.57°	Middle of the drive path.
20 m	0.43°	Typical lane-line range.
30 m	0.29°	Far end of the path. The plan's registration test point.

Parallax for a point at distance *d* is atan(0.150 m ÷ *d*). The table is the plan's own formula evaluated at each distance; see Appendix A.

The eye box is the region where the eye can see the whole image. The plan requires at least 120 × 60 mm without tracking. With tracking, the box follows the eye, so the test span of ±75 mm (150 mm total) is larger than the static box and is only met with tracking on. The plan treats that as intended: the tracker is part of the design, not an optional extra.

5 Self-driving integration

5.1 Why openpilot

openpilot is chosen over other open driving stacks for three reasons. It is the only one with a maintained list of supported cars. It enforces a hardware safety model in panda firmware. And its message bus can be tapped by a second machine without changing the driving code. Alternatives such as Autoware or a research stack would need a drive-by-wire interface built from scratch, and they would carry none of the safety model. The HUD depends on the bus, so the choice of stack determines how much of the HUD can be built on existing code.

5.2 Track A, the critical path: the comma four drives

Track A puts a comma four and a car harness in a supported car and runs the fork's branch on it, so the HUD and the driving computer share one source tree. The driver engages openpilot normally. The HUD mirrors the engagement state. Nothing about openpilot's driving behaviour is modified for the HUD; the HUD only adds a subscriber.

The changes on the comma four are small and are listed below. Each one is a place the plan is most likely to need an adjustment once the hardware is in hand.

ITEM	CHANGE	STATUS
`bridge`, on the comma four	Started onroad. Today `process_config.py` starts it only for `notCar` builds, so a Params flag is added that enables it in a car.	To build
USB-C Ethernet adapter	Links the comma four to the Jetson's RJ45 port with a short Cat6 run. The comma four's USB 3.1 port powers the adapter.	**VERIFY** host mode and USB Ethernet driver
Driving code and panda	Unchanged.	By design
Fallback link	The comma four's Wi-Fi hotspot to the Jetson, accepting the jitter noted in the latency budget.	Only if the adapter fails

Hardware note: the comma 3X is no longer sold. The comma four is $999, or $699 with trade-in, and the harness is $99 (comma shop, read 2026-10-08). The Phase 0 order is the comma four and the harness, and the car is chosen before either is ordered.

5.3 Track B, research: openpilot on the Jetson

Track B is kept alive for a practical reason. JetPack 6.2 (CUDA 12.6, TensorRT 10.3) removes the TensorRT 8.5 parser failure found on 2026-10-07, when the driving model could not be parsed on the Jetson's JetPack 5.1.3 install. Track B is worth doing only if it lets the Jetson run the whole driving stack, not just the HUD. The blockers below were each verified in this checkout.

BLOCKER	WHERE	WHAT IT WOULD TAKE
No CUDA model build	`openpilot/selfdrive/modeld/SConscript` lines 27–31; `helpers.py` , `load_oob`	Add a `DEV=CUDA` (or `NV`) branch and a device case. tinygrad lists both devices, but openpilot never uses them. Correctness on sm_87 is **UNVERIFIED**.
pandad is SPI-only	`openpilot/selfdrive/pandad/panda.cc` line 16	A `PandaUsbHandle` path in C++ for an external red panda, or a Python pandad. Not trivial.
No IMU or GPS on PC	`process_config.py` : `sensord` line 88, `qcomgpsd` line 101, `ubloxd` and `pigeond` lines 105–106 (`timed` at line 83 is also PC-disabled)	A Jetson hardware class, plus sensord for an external IMU and ubloxd for a ZED-F9P. `hardwared` already runs on PC; it is the sensor processes that are missing.
Camera mismatch	`webcam/camera.py` at 1280 × 720; `camera.py` ("pc","unknown")	A GMSL AR0234 kit, the only JetPack 6.2-tested one found, with its own `CameraConfig` and warp file size.

The first blocker is the one that decides Track B. Without a CUDA build of the model, nothing else matters, because the driving model is the part that must run for openpilot to drive. The plan's first milestone is therefore narrow: `modeld` running on CUDA against a replayed route, with outputs matching the Qualcomm build.

> **Open before Track B starts**
> The plan does not define "matching". Before the parity test begins, set a numeric tolerance for the model's outputs (path points, lane-line probabilities and lead distances) and the replay route it is measured on. Without a tolerance, the go/no-go decision has no criterion and can be argued either way.

Go/no-go happens after Track A is driving. Track B does not block the HUD: if it fails, the Jetson remains a HUD computer, and the design does not change. That is the reason Track B sits after week 12 on the schedule (section 9).

6 Optics

The optics are the hardest part of the plan, and the part where the monitors alone will not get the job done. The README commits to projection onto a relay mirror. This chapter designs that path and adds two cheaper stages before it, so software work is never blocked on glass. Each subsection explains the physics before the numbers, and Appendix A derives every formula used.

6.1 The constraints

Six constraints define the optical design. Each is a requirement on what the driver sees, not on a component.

QUANTITY	VALUE	WHAT IT MEANS, AND ITS BASIS
Virtual image distance	≥ 6 m, target 7.5 m	The distance at which graphics appear to sit. Section 2.2 explains why it must be near road distance. Basis: TI DLP application notes; Mercedes and VW use 10 m, Huawei and Hyundai 7.5 m.
Field of view	10° × 4° to 13° × 5°	The angle the window covers. Production AR HUDs fall in this range. The corrected design, 10.8° × 5.4°, is within the width range and slightly above the height range (section 6.2).
Eye box	≥ 120 × 60 mm	The region the eye can move through while still seeing the full image. Head tracking extends it.
Luminance at the eye, noon	≥ 1,500 cd/m²	Needed for the graphic to be visible against the sky. Set from a contrast ratio of 1.15 against 10,000 cd/m² ambient (Analog Devices design note).
Luminance at the eye, night	≤ 50 cd/m²	A ceiling for dark-cabin use, so the overlay does not dominate the scene. Sets the dimming requirement in section 6.5.
Parallax from a 150 mm head shift	0.86° at 10 m, 0.29° at 30 m	Geometry. Head tracking corrects it (section 4.7).

The contrast ratio in the plan is defined as (overlay + background) ÷ background. With a background of 10,000 cd/m² and a ratio of 1.15, the overlay must add 0.15 × 10,000 = 1,500 cd/m², which is where the noon target comes from. The definition matters because a different one, such as the ratio of overlay to background, would give a different target.

6.2 Window geometry

A field of view is an angle, and the physical size of the image depends on the distance it is drawn at. At a virtual image distance of 7.5 m, the window's width is $2 \times 7.5\ \text{m} \times \tan(\theta/2)$, where θ is the field of view. The window's aspect ratio, the width divided by the height, must match the picture generator's aspect ratio. The PGU is 60 × 30 mm, which is 2:1. Figure 10 shows the three candidate windows at true scale.

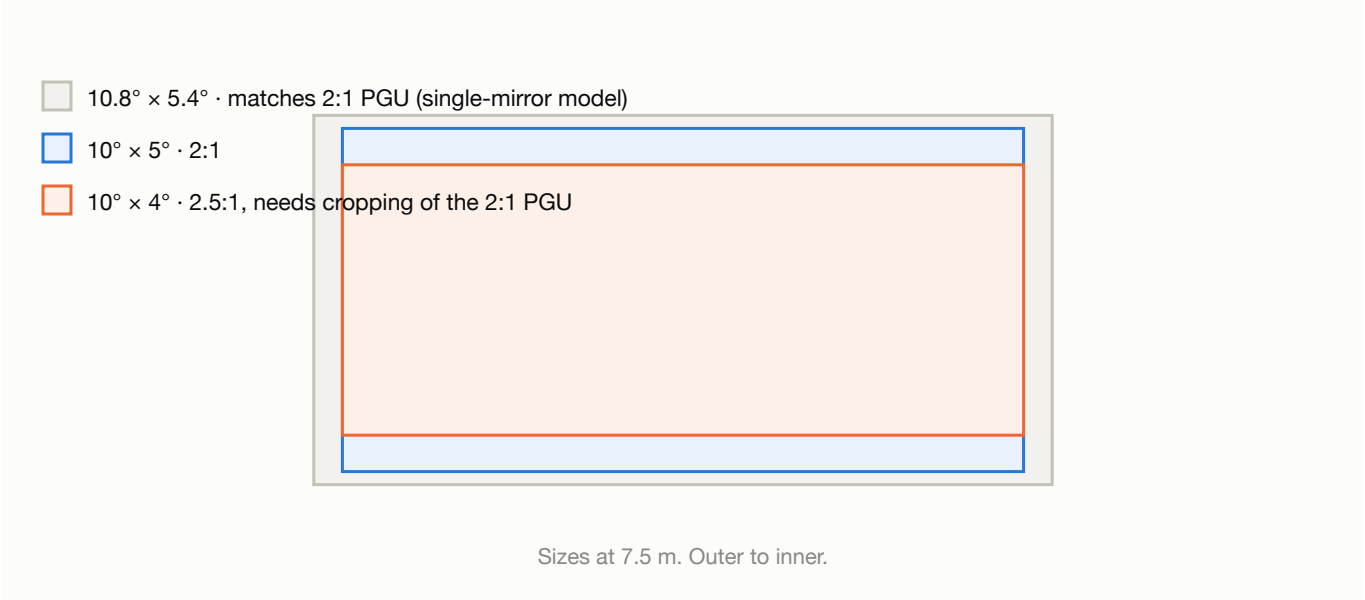


Figure 10. Demo window sizes at a 7.5 m virtual image, drawn to true scale and centred. The 2:1 windows match the PGU directly. The 2.5:1 window needs the PGU's active area cropped.

WINDOW AT 7.5 M	SIZE	MAGNIFICATION NEEDED, 60 × 30 MM PGU
10° × 4° (revision 2)	1.31 × 0.52 m	21.9 across, 17.5 down
10° × 5°	1.31 × 0.65 m	21.9 across, 21.8 down
10.8° × 5.4° (corrected design)	**1.42 m × 0.71 m**	**23.7 both ways**

Revision 2 gave the demo window as 10° × 4°. That is a 2.5:1 window, and the PGU is 2:1. A 2:1 picture cannot fill a 2.5:1 window without losing height, so the 10° × 4° demo would need the PGU's active area cropped to about 55 × 22 mm at the corrected magnification. The corrected design sizes the window to the PGU instead, at 10.8° × 5.4°, and the table shows that the 10° × 5° window needs the same magnification.

6.3 Reflection off glass, and why the source must be s-polarised

When light meets a glass surface, part is transmitted and part is reflected, and the fraction reflected depends on the angle and on the light's polarisation. Polarisation has two components. The *s* component is perpendicular to the plane of incidence; the *p* component is parallel to it. For glass with a refractive index of 1.52, the Fresnel equations give the reflected fraction for each. Figure 11 plots them against the angle of incidence.

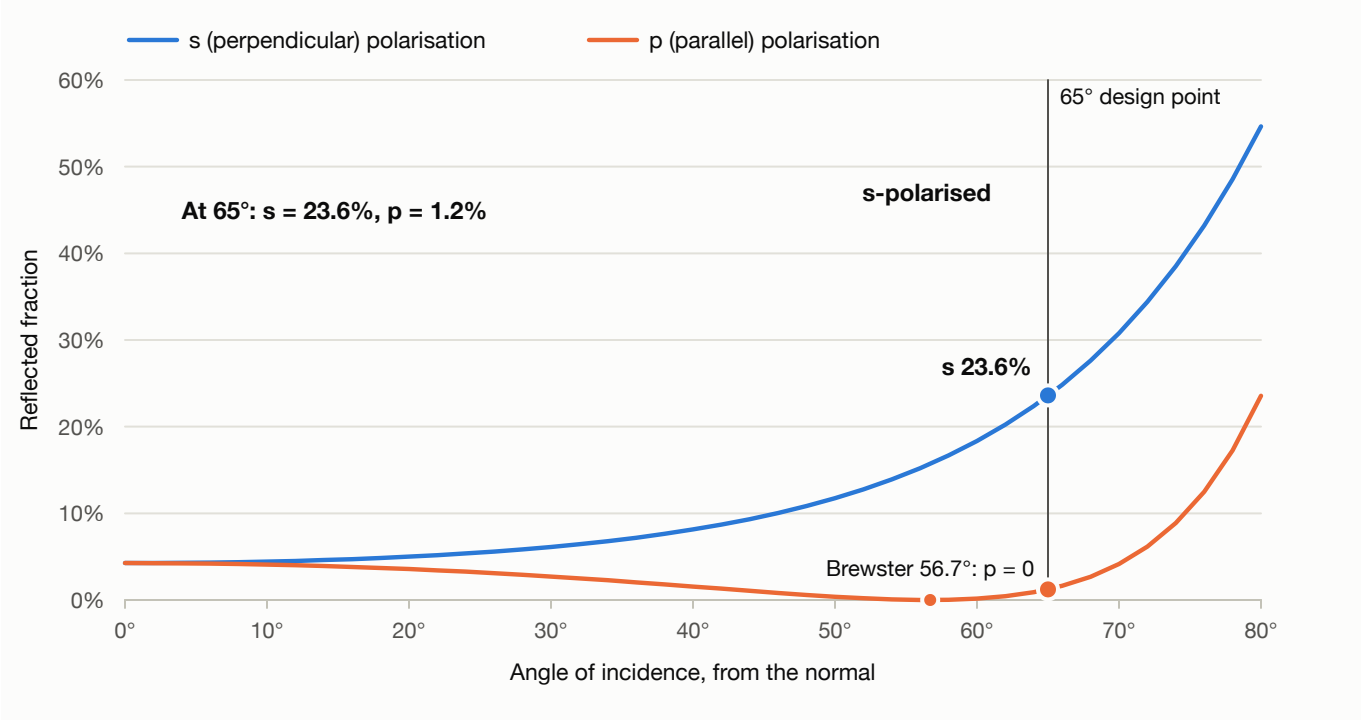


Figure 11. Reflected fraction of a glass surface (n = 1.52) against angle of incidence, for the two polarisations. The p curve reaches zero at the Brewster angle; the design point at 65° is marked.

At the plan's design point of 65°, the s component reflects 23.6% of its light and the p component reflects 1.2%. The difference is the design decision. The windshield is a reflector for the HUD, so the picture has to be s-polarised to be bright, or the projector's output has to be converted to s-polarisation with a film. The laser DLP projector's output may already be polarised; its polarisation has to be measured before the film is ordered. Revision 2 said the same thing without showing the curve.

Two consequences of the curve were not in revision 2.

- **The monitor is already polarised.** An LCD panel emits linearly polarised light through its front polariser. The orientation in which the monitor is mounted therefore decides whether the windshield sees s-polarised light and reflects 23.6% of it, or p-polarised light and reflects 1.2%: a twenty-fold difference in brightness from a rotation of the panel. Before the monitor is used as the night picture generator in Stage 2, its polarisation axis must be measured against the plane of incidence, and the panel rotated or a half-wave retarder film added so the light arrives s-polarised. The Stage 0 and Stage 1 combiners are coated beamsplitters whose reflection can also depend on polarisation, and the plan has no figures for them (`VERIFY VENDOR`), so the same orientation check is done there first.
- **The diffuser scrambles the projector's polarisation.** A rear-projection diffuser depolarises the light that passes through it, so whatever the laser DLP emits, the light leaving the PGU is close to unpolarised. The polariser sheet in the parts list is therefore needed, not optional, and it passes only about 42% of unpolarised light (an assumed figure for an iodine film polariser; a wire-grid polariser passes a little more). The budget in section 6.5 now carries that loss as its last row.

The angle of 65° is the plan's design incidence on the windshield. The real angle depends on the eye point and the mirror position, and the ray trace in Stage 2 (section 6.6) will confirm it. If the real angle differs a great deal, the reflected fraction changes with it, and the luminance budget in section 6.5 must be recomputed.

6.4 The ghost image and the wedge interlayer

A laminated windshield has two reflecting surfaces, the outer and the inner, separated by the plastic interlayer. Each reflects the picture, so the driver sees two images. In a plain laminate the two surfaces are parallel, so the two images are displaced sideways by a fixed amount. That is the ghost image. A wedge-shaped interlayer tilts the inner surface by a small angle, and the two reflections then leave the glass at different angles. Choosing the wedge so that the two reflections meet at the design eye point makes the ghost disappear at that point.

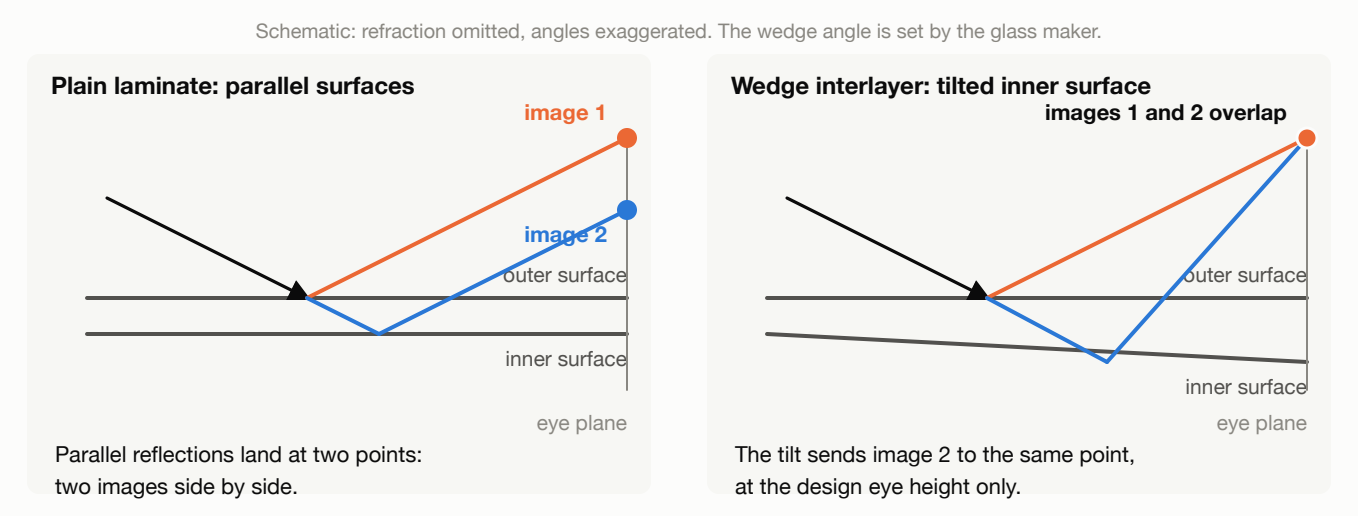


Figure 12. Two reflections from a laminated windshield. In a plain laminate the two images are parallel and separate; the wedge tilts the inner surface so that, at the design eye point, the two images overlap. Schematic only.

The wedge removes the ghost only at the nominal eye height. Above or below it, the two images separate again, and the driver sees a double image when the head moves. For this reason the plan treats the wedge as the fix for the nominal case and the eye-box test as the check for the rest. The wedge angle is set by the glass maker, and this plan does not quote one. That is also why the plan favours a car that has a factory head-up display trim: a wedge-PVB windshield for that trim is more likely to exist.

6.5 The luminance budget

Luminance, measured in cd/m² or nits, is the brightness of a surface as seen by the eye. It is conserved by an ideal imaging system: a mirror that magnifies an image makes it bigger, not brighter. So the budget is a chain of multipliers. Each reflection or transmission multiplies the source luminance by a fraction, and the result is the luminance at the eye. The source luminance of a laser DLP is computed from its light output and the diffuser's area. For a Lambertian transmissive diffuser that is $L = \Phi\eta \div (\pi A)$, where Φ is the projector's output in lumens, η is the diffuser's transmission, and A is its area. For 2,000 lm, $\eta = 50\%$, and a 60 × 30 mm diffuser, that gives 176,839 cd/m² at the source.

SOURCE AND PATH	AT SOURCE	COMBINER PATH (×0.30)	WINDSHIELD PATH (×0.24)	VERDICT
YODOIT 16″ 4K, peak	1,200	360	288	Night, dusk, overcast only. About 19% of the noon target.

SOURCE AND PATH	AT SOURCE	COMBINER PATH (×0.30)	WINDSHIELD PATH (×0.24)	VERDICT
YODOIT 16″ 4K, sustained **UNVERIFIED**	~600	180	144	Night only.
Laser DLP, 2,000 lm on the 60 × 30 mm diffuser, 50% efficient	176,839	53,052	42,441	Daylight, about 28× the noon target. Needs dimming for night.
Laser DLP, windshield path with two relay mirrors at 90% each **ASSUMED**	176,839	n/a	34,377	Daylight, about 23× the noon target, if the light reaches the windshield s-polarised without a polariser.
Laser DLP, same path with a 42% polariser sheet after the diffuser **ASSUMED**	176,839	n/a	14,439	Daylight, about 10× the noon target. This is the realistic Stage 2 figure (section 6.3).

Values in cd/m². The two path columns are alternatives, not a cascade: the combiner path does not pass through the windshield. The windshield path has two relay mirrors in addition to the windshield, and the last row adds the polariser sheet. The 90% reflectance and the 42% polariser transmission are assumptions; the plan has not measured either. The 0.24 windshield factor is for one surface and is conservative (Appendix A.5).

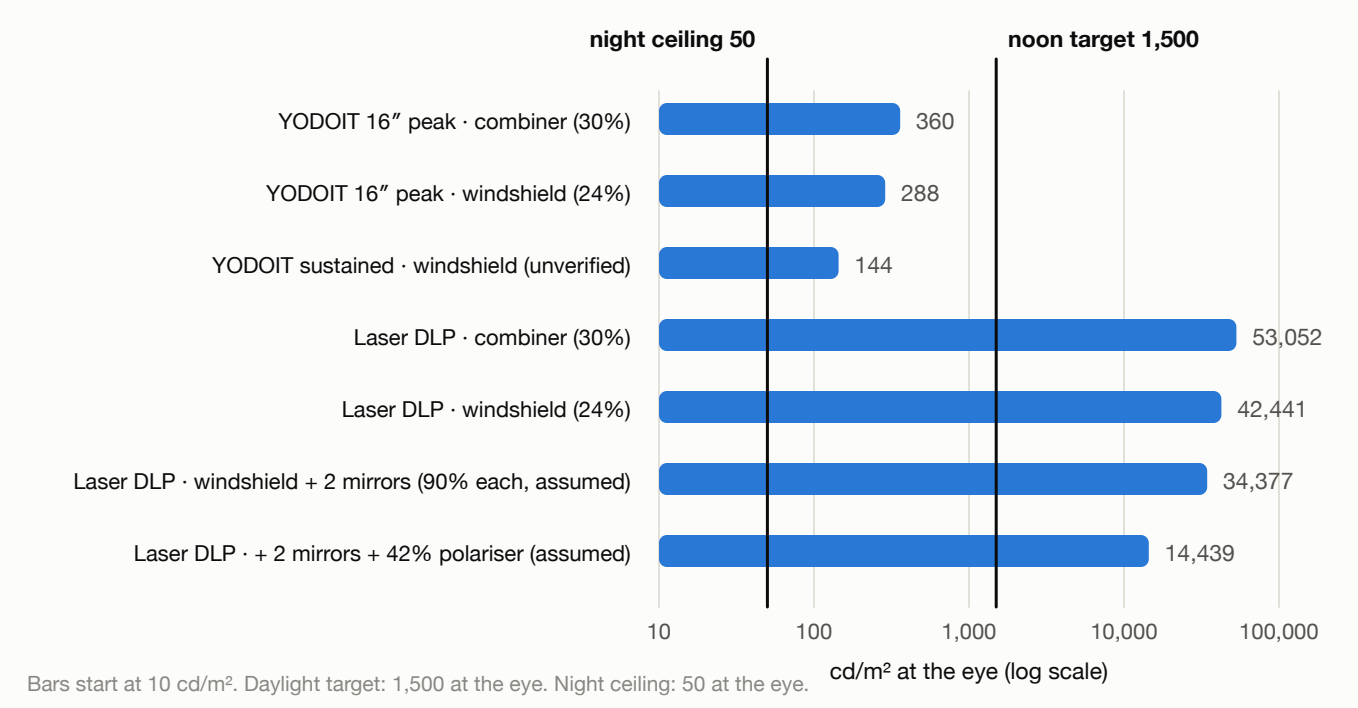


Figure 13. Luminance at the eye for each source and path, on a log axis, against the noon target (1,500) and the night ceiling (50). Solid lines mark thresholds; the bars run from the axis minimum of 10.

The dimming requirement follows from the two thresholds. From the 1,500 cd/m² noon target down to the 50 cd/m² night ceiling is a ratio of 30:1, which is what the revision 2 plan should have said. If the projector runs at its full output, 42,441 cd/m² at the eye, the range from full output down to 50 cd/m² is 849:1. The daylight target needs only about 10% of the projector's full output on the realistic path with mirrors and polariser (1,500 ÷ 14,439), and 3.5% on the bare windshield path, so the realistic working range is the first figure. Whether the projector can be dimmed that far is not known: many laser projectors do not dim below roughly

a fifth of full output by source control alone, and the floor of the chosen model must be measured (VERIFY). If it cannot reach the daylight working point, a fixed neutral-density filter sets that point and the switchable filter takes it to night. The plan specifies projector dimming plus a switchable neutral-density filter for night.

The monitors do not reach the daylight target, at about 19% of it through the windshield path, which is why they are used only for the bench rig and for night driving. The laser DLP reaches daylight with headroom, which is why it is the Stage 2 picture generator, and why the sun hazard in section 6.7 matters so much.

6.6 Three build stages

The optics are built in three stages. Each stage tests a piece of the final design, and the first two can be built before the custom mirror arrives. Stage 0 is a bench rig that tests software on a desk. Stage 1 places a combiner in the car and tests registration at a short virtual distance. Stage 2 adds the relay mirror and the windshield and is the demo target.

Stage 0: bench rig (week 1)

The monitor sits face-up in a black enclosure, and a 70/30 teleprompter beamsplitter is set at 45° above it. A camera, or the developer's eye, looks through the beamsplitter from the far side. The flat beamsplitter forms a virtual image of the monitor at the same distance as the monitor, about 0.6 m. Teleprompter glass uses a convention in which the first number is the transmitted share, so 30% of the monitor's light is reflected toward the viewer. That reading is an interpretation of the vendor's convention and should be confirmed with the vendor (VERIFY VENDOR).

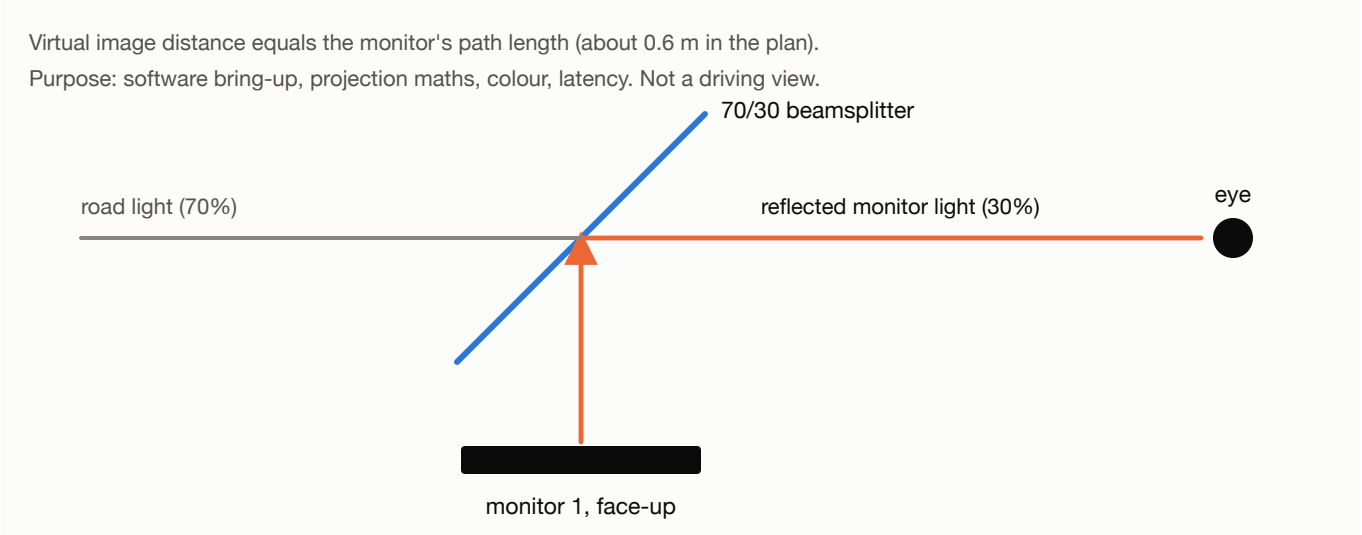


Figure 14. Stage 0 bench rig. The monitor's light folds 90° off the beamsplitter to the eye. Schematic.

The purpose of Stage 0 is software: projection maths, colour and alpha choices, and latency instrumentation. It has no virtual distance and no windshield, so it tests the renderer rather than the optics. The plan notes that teleprompter glass is sold in sizes of 300 mm and above by teleprompter suppliers, while the optics catalogues stop near 85 mm (VERIFY VENDOR).

Stage 1: Fresnel magnifier and in-car combiner (weeks 4–8)

Stage 1 moves the rig into the car, on the dash top. A 300 mm Fresnel lens sits between the monitor and a combiner, a free-standing glass above the dash. The monitor is 280 mm from the lens, which is inside the 300 mm focal length. An object inside the focal length of a converging lens produces a *virtual* image on the same side as the object, magnified. That is how a magnifying glass works, and it is how Stage 1 makes a virtual image.

```
1/v = 1/f – 1/u = 1/300 – 1/280  so  v = –4.2 m   (virtual, on the monitor side)
magnification M = |v| / u = 4200 / 280 = 15
image = 80 × 40 mm × 15 = 1.2 × 0.6 m  so  16° × 8° as seen from the eye
pixels: 80 mm of a 277 ppi panel ≈ 872 px; 872 px ÷ 16° ≈ 54 px per degree
```

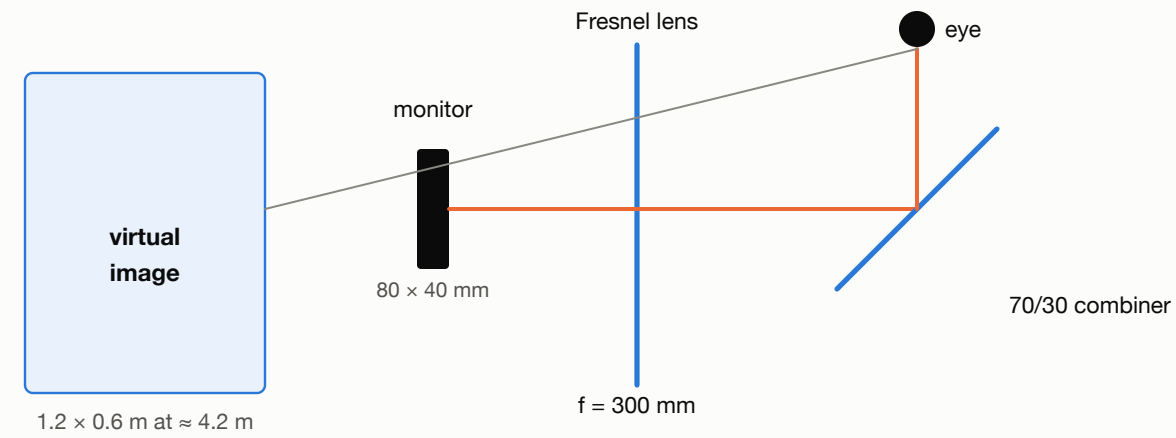


Figure 15. Stage 1 geometry. The monitor sits inside the lens's focal length, so the lens makes a virtual image about 4.2 m away on the monitor side. The combiner folds the light toward the eye. Not to scale; the derivation is in Appendix A.

Fifty-four pixels per degree is close to the eye's limit of about 60 for 20/20 vision, so the Stage 1 image is sharp enough to judge registration. Fresnel lenses add ring artefacts and some colour fringing, which is acceptable for a first in-car result and for night demos. Stage 1 leaves the windshield untouched, which keeps the legal question simpler (section 11). The combiner is more exposed than the Stage 2 housing, because it stands above the dash.

Stage 2: relay mirror into the windshield (weeks 6–12)

Stage 2 is the README's design and the demo target. The light path is PGU, then a fold mirror, then a concave mirror, then the windshield, then the eye. The concave mirror does the magnification; the windshield is the final reflector and sits between the mirror and the driver.

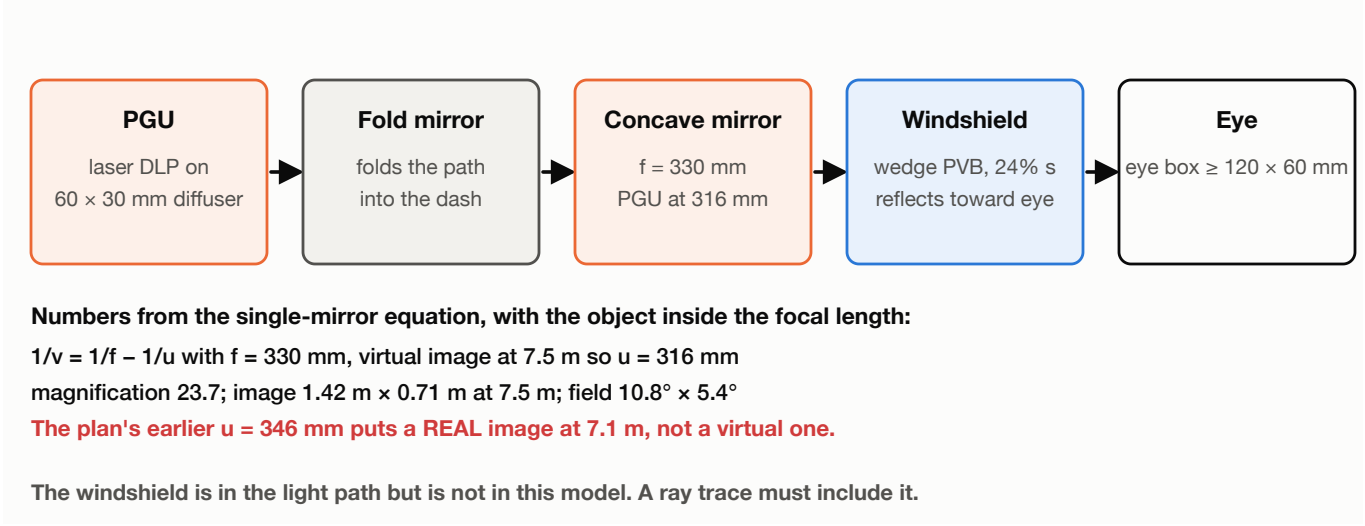


Figure 16. Stage 2 light path, with the single-mirror numbers. The windshield is drawn in the path but is not part of the calculation, which is the open issue described below.

The single-mirror calculation uses the mirror equation, 1/v = 1/f – 1/u, with f = 330 mm. For a virtual image the object must lie inside the focal length. Setting the virtual image at 7.5 m gives an object distance of 316 mm, a magnification of 23.7, an image of 1.42 m × 0.71 m at 7.5 m, and a field of 10.8° × 5.4°.

> **Correction: the revision 2 geometry put the image in the wrong place**
> Revision 2 used u = 346 mm with f ≈ 330 mm. Since 346 mm is outside the focal length, the same equation gives a *real* image 7.1 m from the mirror, not a virtual one. The magnification of about 22 that revision 2 quoted matches a real image about 7.5 m from the mirror, which is not the virtual image the design needs. A real image cannot be seen through the windshield as a virtual image at 7.5 m, so the earlier numbers are withdrawn and replaced by the calculation above.

The single-mirror calculation is still not enough to order a mirror. The windshield is a second reflecting surface between the concave mirror and the eye. Its curvature and its position change where the eye perceives the image, and neither is in the calculation. The ray trace that fixes f, u and the mirror's shape must include the windshield's actual curvature for the chosen car, the eye point, and the eye box. Until that ray trace exists, the mirror is a design estimate.

> **Sequencing consequence**
> The windshield's curvature depends on the car, so the car decision must come before the mirror is specified. Revision 2 scheduled the mirror order for week 1, in the same phase as the car choice. The mirror order should move to after the ray trace, which needs the car. Section 9 shows the consequence for the schedule.

Two further facts about the mirror come from the plan. It is a custom spherical first-surface mirror, about 300 × 200 mm, quoted by an optics shop; for a spherical mirror the radius of curvature is twice the focal length, so about 660 mm; no catalogue mirror is that size. A spherical mirror used off-axis has astigmatism and

keystone, a distortion in which the image's top and bottom have different widths. The keystone is corrected in software, since hudd pre-distorts the image, and the residual astigmatism is accepted for the demo. A freeform mirror from an automotive HUD supplier is the stretch option.

6.7 The sun-focus hazard

The concave mirror is a converging reflector. Sunlight that enters through the windshield and strikes the mirror is focused toward the mirror's focal point, the same way a magnifying glass focuses the sun onto paper. For the Stage 2 geometry, that focal point is 330 mm from the mirror, and the PGU sits at 316 mm. The diffuser is therefore 14 mm short of the focal point, close enough that a focused spot falls on or near it. The plan's risk register names this, and this section quantifies it.

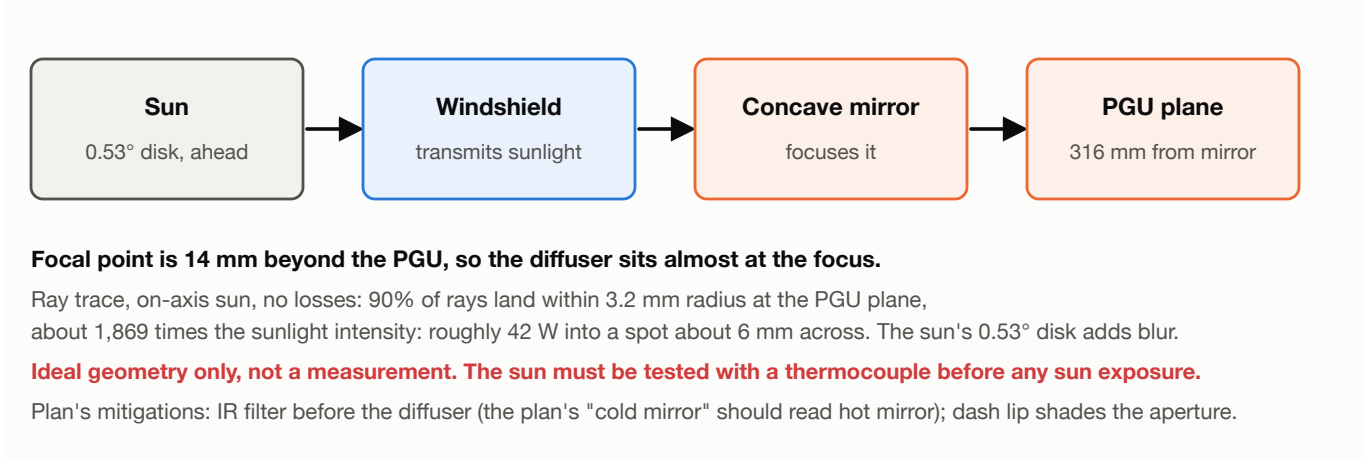


Figure 17. The sun-focus path and the geometric estimate. The estimate is an upper bound: it assumes an on-axis sun, parallel rays, and no losses.

The estimate used a ray trace of parallel rays on a 300 × 200 mm spherical mirror of radius 660 mm. At the PGU plane, 90% of the rays land within 3.2 mm of the axis. The sun's own angular size, 0.53°, adds blur: at the focal plane, its image is about 3.1 mm across. Taking the mirror's aperture against the 90% spot gives a concentration of about 1,869 times the sunlight intensity. Since the trace has no windshield losses and no sun disk, that number is an upper bound. The real value is lower, because of windshield losses and the sun's disk, but the trace does not say by how much. The conclusion does not depend on that number: the diffuser would see sunlight concentrated far beyond ordinary exposure.

In power terms, the mirror's 300 × 200 mm aperture collects about 42 W of direct sunlight at 1,000 W/m², after an assumed 70% windshield transmission, and the trace puts most of it inside a spot about 6 mm across at the PGU plane. That is the power of a small soldering iron delivered to a patch of plastic film, and it is why the thermocouple test below is an acceptance test and not a precaution.

Two points in the plan need correcting before the hazard is managed:

- **Terminology.** The plan asks for a "cold mirror" in front of the diffuser. A cold mirror reflects visible light and transmits infrared. Keeping infrared off the diffuser needs the opposite: a hot mirror, which reflects infrared and passes visible light, or an IR-cut filter. Confirm the term with the optics vendor before ordering.
- **Filters reduce heat; they do not remove the focus.** A filter lowers the heat load on the diffuser. It does not stop visible sunlight from converging at the PGU. The plan's shading of the housing aperture by the

dash lip is likely to help only at some sun positions; the plan does not quantify it.

> **Proposed for review, not in the plan**
> Keep sunlight from entering the mirror's aperture at low sun, for example with a shutter or a shade that the driver cannot override, and measure the diffuser's temperature with a thermocouple in the sun before any sun exposure of the assembled Stage 2 unit. The thermocouple test is the acceptance test; the shutter is a proposal.

6.8 Packaging

The whole optical train sits in a 3D-printed, matte-black housing on the dash top. The mirror aperture is under a dust-cover glass at the base of the windshield. The housing comes off without tools so the car can return to stock, which matters for the vehicle modification sign-off (section 11). Matte black inside the housing is meant to limit stray light, which would otherwise appear as haze or a faint second image.

6.9 Calibration

Three transforms take a point from the car's frame to a pixel on the panel. Each is measured once and stored in Params on the Jetson, except the first, which the head tracker adjusts live. Figure 18 shows the chain and how each step is measured.

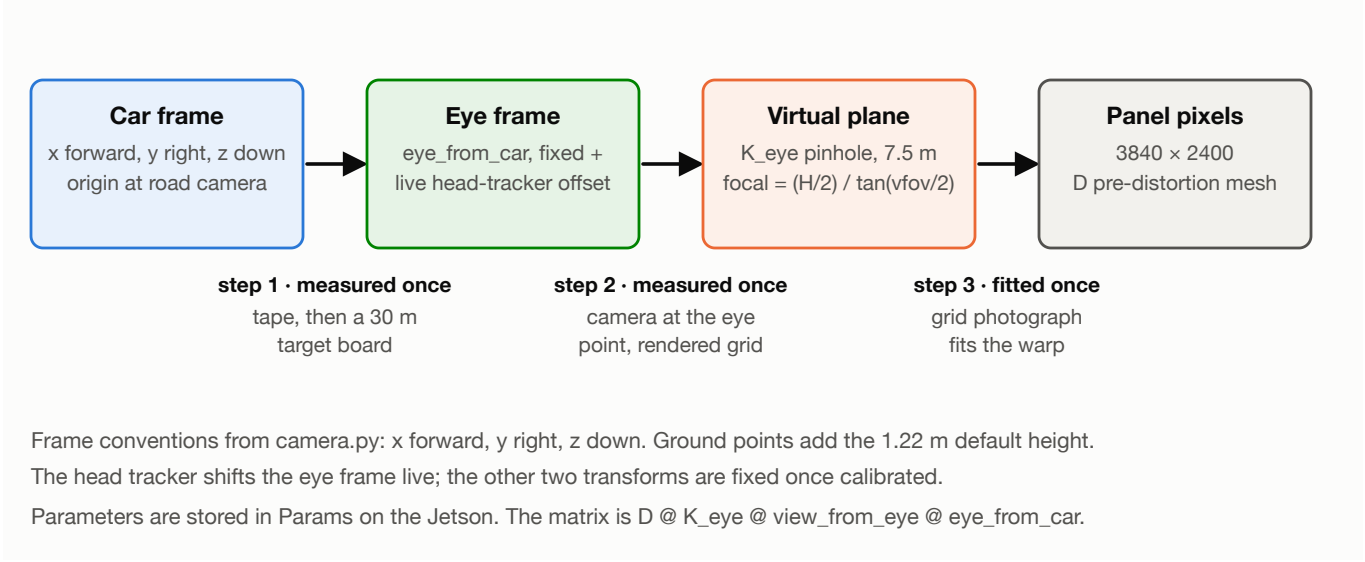


Figure 18. The projection chain from the car frame to the panel, with each transform's measurement method. The matrix is D @ K_eye @ view_from_eye @ eye_from_car, applied in that order to a point.

1. **Car frame to eye.** A fixed translation from the comma four's camera origin to the nominal eye point. It is measured with a tape and refined with a target board at 30 m. The head tracker supplies a live offset from the nominal point.
2. **Eye to virtual display plane.** A pinhole intrinsic for the virtual image. Its focal length in pixels is half the display height divided by the tangent of half the vertical field of view. It is measured by putting a camera at the eye point and photographing a rendered grid against a real grid on the ground.

3. **Display pre-distortion.** A two-dimensional mesh warp that cancels the mirror's and windshield's curvature. It is fitted from the same photograph and applied as a final full-screen pass in the renderer, not folded into the matrix.

The calibration depends on the windshield, so it must be redone for every car and every windshield change. It is part of the Stage 2 re-calibration in phase 5 (section 9).

7 Software

All new code lives in this fork under `openpilot/selfdrive/hud/` and reuses openpilot's messaging, Params and UI framework. The existing UI is left as it is, so the comma four's own screen is unchanged. The new code falls into five processes, a projection chain, and a set of bring-up facts that were checked in the source.

7.1 New processes on the Jetson

Each job is a separate process. A fault in one does not take down the others, and each can be tested alone on the bench. The plan does not yet state a restart policy for a crashed process; that is an open item for Phase 1.

PROCESS	ROLE	BUILT FROM
`bridge <comma-ip> <whitelist>`	Subscribes to the comma four's ZMQ ports and republishes locally as msgq. The whitelist is a substring match in `bridge.cc` line 13, so use exact, full service names (section 4.3).	Existing binary, `openpilot/cereal/messaging/bridge.cc`
`hudd`	The HUD renderer. A `SubMaster` on `modelV2`, `radarState`, `carState`, `controlsState`, `selfdriveState`, `extrinsicsCalibration`, `deviceState`, `pandaStates` and `longitudinalPlan`, plus the two local services below. Projects into the eye frame, predicts by age, pre-distorts, and draws at 4K60 full screen on the HUD output.	New. Reuses `ModelRenderer` with a new transform from `set_transform`. Replaces the video-crop matrix in `augmented_road_view.py` with an eye-camera matrix.
`headtrackd`	Runs MediaPipe Face Landmarker on the Jetson against the OAK-D Pro's stream, with the camera's IR lighting for night. Running a converted landmark model on the camera's own processor instead is an option to **VERIFY**. Publishes a new `hudEyePose` service at camera rate, with a 3D eye position in the car frame and a confidence value.	New service in `log.capnp` and `services.py`
`perceptd`	Runs a TensorRT 10 detector (signs, pedestrians, cyclists) on a road camera attached to the Jetson. Publishes `hudDetections` with 3D positions from a ground-plane assumption. HUD-only cues.	New. Camera: the AR0234 GMSL kit, or a USB global-shutter camera until it arrives.
`hudmirror`	Composites the rendered frame, the driver camera with the eye point, and the latency and registration numbers for the audience on monitor 2.	New, small, built on pyray.

The two new services matter for the rest of the stack. `hudEyePose` is the only input that changes the projection on every frame, and `hudDetections` is the only input that carries something openpilot does not produce. Both are published from the Jetson, and neither is consumed by openpilot, so neither can change how the car is driven.

7.2 The projection, concretely

Today's UI computes `video_transform @ K_cam @ view_from_calib` and passes the result to `ModelRenderer.set_transform`. That maps the model's points onto the camera's video. The HUD replaces it with a chain that maps the same points onto the virtual image:

```
D @ K_eye @ view_from_eye @ eye_from_car
```

- `eye_from_car` is the rigid transform from the calibrated model frame (x forward, y right, z down, origin at the road camera) to the eye. The head tracker's live offset is added here. Points on the ground still need the camera height, `extrinsicsCalibration.height[0]`, with a default of 1.22 m, added in z. `model_renderer.py` does this today.
- `K_eye` is the pinhole for the virtual display plane, from the calibration in section 6.9.
- `D` is the pre-distortion mesh. It is applied as a render-to-texture pass, not folded into the matrix, because a mesh cannot be expressed as a single matrix.

Figure 18 in the optics chapter shows the same chain as boxes with the measurement method for each step. The age compensation of section 4.6 is applied to the points before this chain runs, so the prediction and the projection are separate, testable steps.

7.3 Drawing limits

`MIN_DRAW_DISTANCE` rises from 10 m to the bottom edge of the virtual window. Points nearer than that would project below the window, where the hood hides the road, so they are not drawn. This keeps the path from appearing to float over the bonnet.

7.4 Bring-up details already known

The following facts were read in the source. Each one changes what must be built first.

- **Full screen on a chosen output.** `system/ui/lib/application.py` never calls raylib's full-screen or monitor functions. hudd adds `set_window_monitor` and `toggle_fullscreen`, and sets `BIG=1`, `FPS=60` and `ENABLE_VSYNC=1`. Vsync stops tearing in the mirror image.
- **GL version.** The shaders are GLES 3.0 or GL 3.3 core. The Jetson's driver supports both, but this is untested in this tree. A sensible first test is a full-resolution triangle, which separates GL problems from renderer problems.
- **Params and state the UI reads locally.** `ui_state.py` reads `IsMetric`, `ExperimentalMode` and `CarParamsPersistent` from local Params, and derives `started` from `deviceState` and `pandaStates`. hudd receives `deviceState` and `pandaStates` through the bridge and copies the few Params it needs once at start, so the HUD agrees with the comma four about units and engagement.
- **Jetson runtime.** JetPack 6.2 gives Ubuntu 22.04 and a modern glibc. That should let openpilot's manylinux_2_28 aarch64 wheels and the uv-managed Python 3.12 install natively. Building the UI alone does not need LLVM. **UNVERIFIED** until installed on the device.
- **Power mode.** Flash with Seeed's guide and keep `nvpmodel` at MAXN, not MAXN SUPER, which Seeed says this carrier cannot cool.

7.5 Bench-first development

Nothing in the software plan waits for a car. Every stage can run on a mini PC with recorded data. `tools/replay/replay --demo` publishes a real route's `modelV2`, `radarState`, `carState` and calibration as msgq. A bridge on the mini PC republishes them to the Jetson exactly as the comma four will. `tools/clip` renders the HUD to MP4 for regression checks. `tools/sim` with MetaDrive exercises engagement-state changes. The Stage 0 rig sits on the same desk.

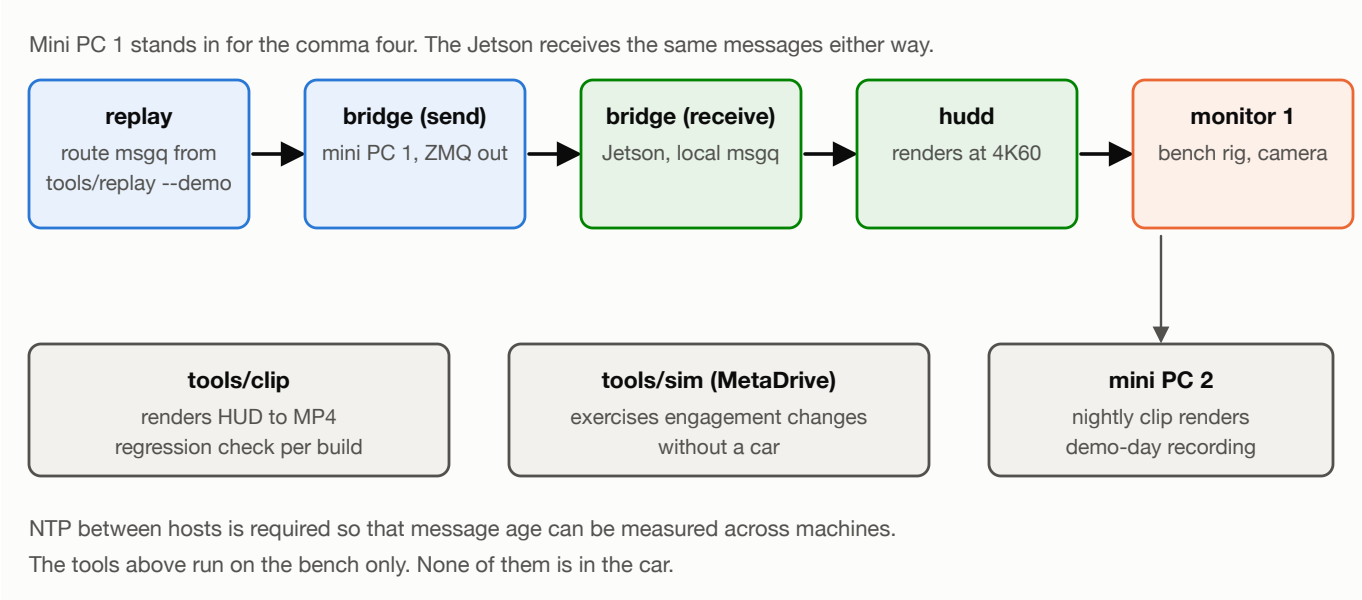


Figure 19. The bench chain. Replayed messages leave mini PC 1 by bridge and arrive on the Jetson in the same form they would from the comma four. The regression tools run alongside and do not touch the car.

Two points keep the bench honest. Replayed data has no live latency, so the bench tests timing only through the bridge and the render loop; the photodiode test in section 10 is the only measure of the real pipeline. And the mini PC is an x86 machine, so it tests logic but not the Jetson's GL driver or its power limits. Bench success is necessary for the Phase 2 gate, not sufficient.

8 Hardware roles and acquisitions

This chapter lists what is already owned, what must be bought, and in what order. The order matters more than usual here, because the two longest-lead parts depend on a decision that has not been made yet.

8.1 What you already have

ITEM	ROLE IN THIS PLAN
Seeed reComputer J4012, Orin NX 16 GB	HUD computer: `hudd`, `headtrackd`, `perceptd`, `hudmirror`. Reflash to JetPack 6.2. Seeed lists support for JetPack 6.0, 6.1 and 6.2 on this carrier. Input is 9–19 V, so it runs from the car's 12 V bus. Run at MAXN, the 25 W class.
YODOIT 16″ 4K 1,200-nit monitor, unit 1	Picture generator for Stage 0 and Stage 1, and the night PGU in Stage 2. 3840 × 2400, 277 ppi.
YODOIT 16″ 4K monitor, unit 2	Audience and debug display on the passenger side, driven by `hudmirror`. Also the spare PGU.
ACEMAGIC mini PC, unit 1	x86 development and replay bench. Runs replay, sim and bridge to stand in for the comma four. Builds the fork.
ACEMAGIC mini PC, unit 2	Demo-day recording and CI box. Renders clip regressions nightly and records the mirror feed and the audience view. Spare for unit 1.

The Jetson is recorded in memory as a JetPack 5.1.3 system with a 3840 × 2400 panel on HDMI-0 and 15 W mode. It has no camera and no panda. Those are the facts of 2026-10-07. The board was unreachable when this plan was written, so they must be checked before flashing.

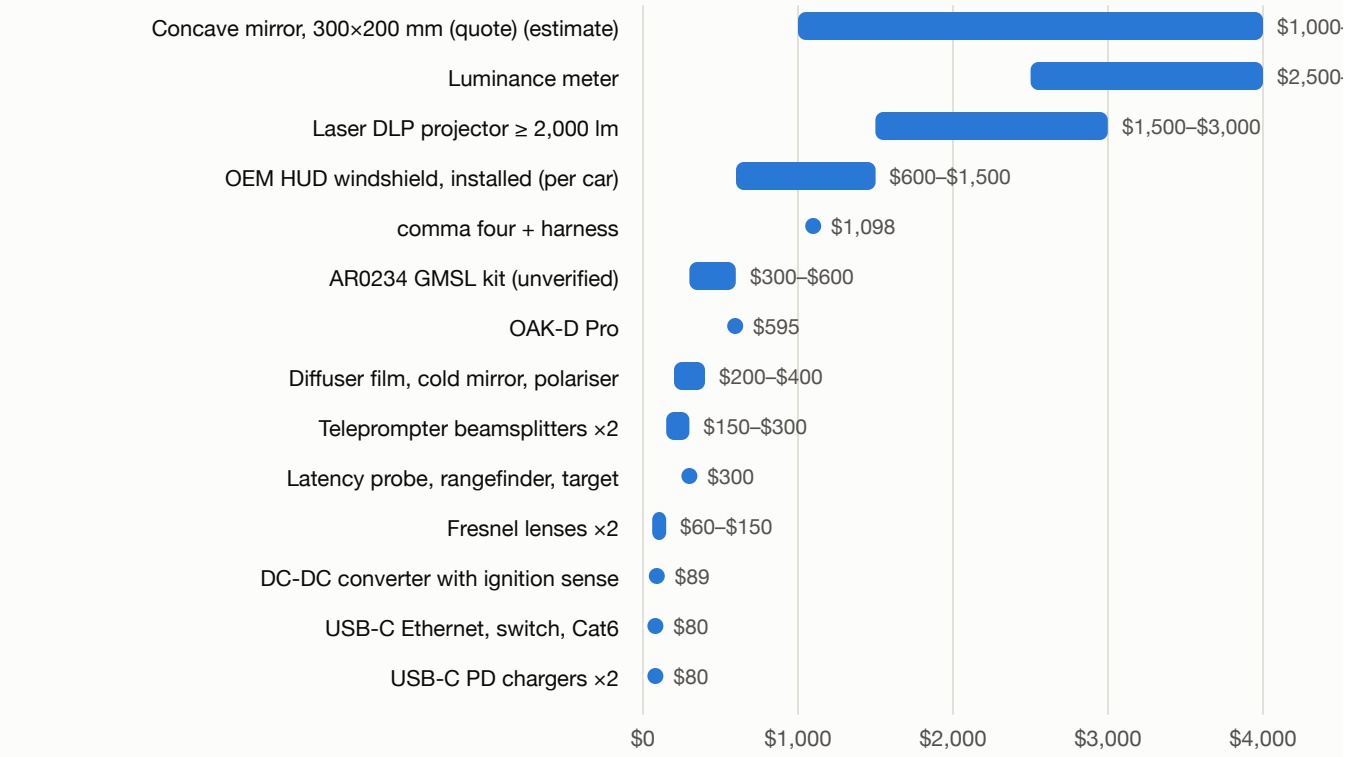
8.2 To acquire

Each row gives the cost and the reason the plan chose this part over a cheaper one. The cost chart shows the same items as ranges. Where the plan gives one price, the bar is a point.

ITEM	APPROX. COST	WHY THIS AND NOT THE CHEAPER THING
comma four and car harness	$1,098	The only configuration in which this tree's pandad, cameras, IMU and GPS all work unmodified. The 3X is no longer sold.
OEM HUD-option windshield for the chosen car, installed	$600–1,500 **PER CAR**	Wedge PVB removes the ghost image, which nothing applied to a plain windshield does. Pick the car so this exists.
Short-throw laser DLP projector, ≥ 2,000 lm	$1,500–3,000	The only source class with daylight headroom after a 24% windshield reflection. The monitors reach about a fifth of the daylight target.
Custom spherical first-surface concave mirror, about 300 × 200 mm, f ≈ 330 mm	quote, likely $1,000–4,000	Sets the 7.5 m virtual image. Order only after the ray trace (section 6.6). Catalogue concave mirrors stop near 100 mm.

ITEM	APPROX. COST	WHY THIS AND NOT THE CHEAPER THING
Rear-projection diffuser film, cold mirror, linear polariser sheet	$200–400	PGU screen, sunlight protection and s-polarisation for the windshield. The polariser is needed because the diffuser depolarises the light, and it costs about half of it (section 6.5). Check the "cold mirror" term (section 6.7).
Teleprompter beamsplitter glass 70/30, 300 × 400 mm, ×2	$150–300	Stage 0 and Stage 1 combiner. One spare for the car.
Fresnel lens 300 mm, f ≈ 300 mm, ×2	$60–150	Stage 1 virtual image at about 4 m before the custom mirror arrives.
Luxonis OAK-D Pro	$595	Global shutter, onboard IR illumination for night head tracking, and a stereo depth output for the eye's distance. MediaPipe Face Landmarker runs on the Jetson; running a converted model on the camera's own processor is an option to **VERIFY**. Tobii and Smart Eye kits are OEM-only.
Arducam AR0234 GMSL kit with deserialiser	$300–600 **UNVERIFIED**	The only JetPack 6.2-tested GMSL camera kit found for Orin NX. Feeds `perceptd` and Track B. Compatibility with the classic J401 carrier must be confirmed with Seeed before ordering.
Mini-Box DCDC-USB-200	$89	Ignition-sensed 12 V supply with clean shutdown for the Jetson. A bare converter would corrupt the filesystem at key-off.
USB-C Ethernet adapter, gigabit switch, Cat6	$80	Wired link for the bridge. Wi-Fi jitter shows up as path wobble.
USB-C PD 100 W car charger, ×2	$80	Monitor power. 4K at full brightness exceeds one USB-C port on some units.
Luminance meter, spot, e.g. Konica Minolta LS-150 class	$2,500–4,000	Needed to prove the contrast criterion. A lux meter cannot measure a virtual image.
Photodiode and microcontroller latency probe, laser rangefinder, 30 m target board	$300	Latency and registration measurements in section 10.
SparkFun ZED-F9P GNSS	$260	Only for Track B or future navigation. The comma four already has GPS. Deferred.

Prices are last seen, not quotes. The two custom optical parts, the mirror and the windshield, have the longest lead times, and the plan's order sequence (section 8.4) changes when they are placed.



Total of the listed ranges: $8,552–$16,192 (before tax, shipping and the deferred ZED-F9P GNSS at $260).

Prices are as listed in the plan (last seen, not quotes). 'estimate' and 'unverified' are the plan's own tags.

Figure 20. Acquisition cost ranges per item, from the table above. Dots are single prices. The total excludes the deferred GNSS receiver and is a sum of the listed ranges, not a quote.

The two largest uncertainties are the mirror, which is a quote and not a price, and the luminance meter, which is needed for the contrast criterion and cannot be replaced by a lux meter. The mirror cost could move the total by several thousand dollars; the meter is a fixed cost that the validation depends on.

8.3 Power and wiring

The Jetson runs from the car's 12 V bus through an ignition-sensed DC-DC converter. Ignition sensing lets the Jetson see the key-off signal and shut down cleanly. Without that, the Jetson loses power mid-write and can corrupt its filesystem, which is the reason the plan lists the Mini-Box as a required part and not an optional one. The monitors take power from two USB-C PD car chargers. The comma four stays on its own car harness.

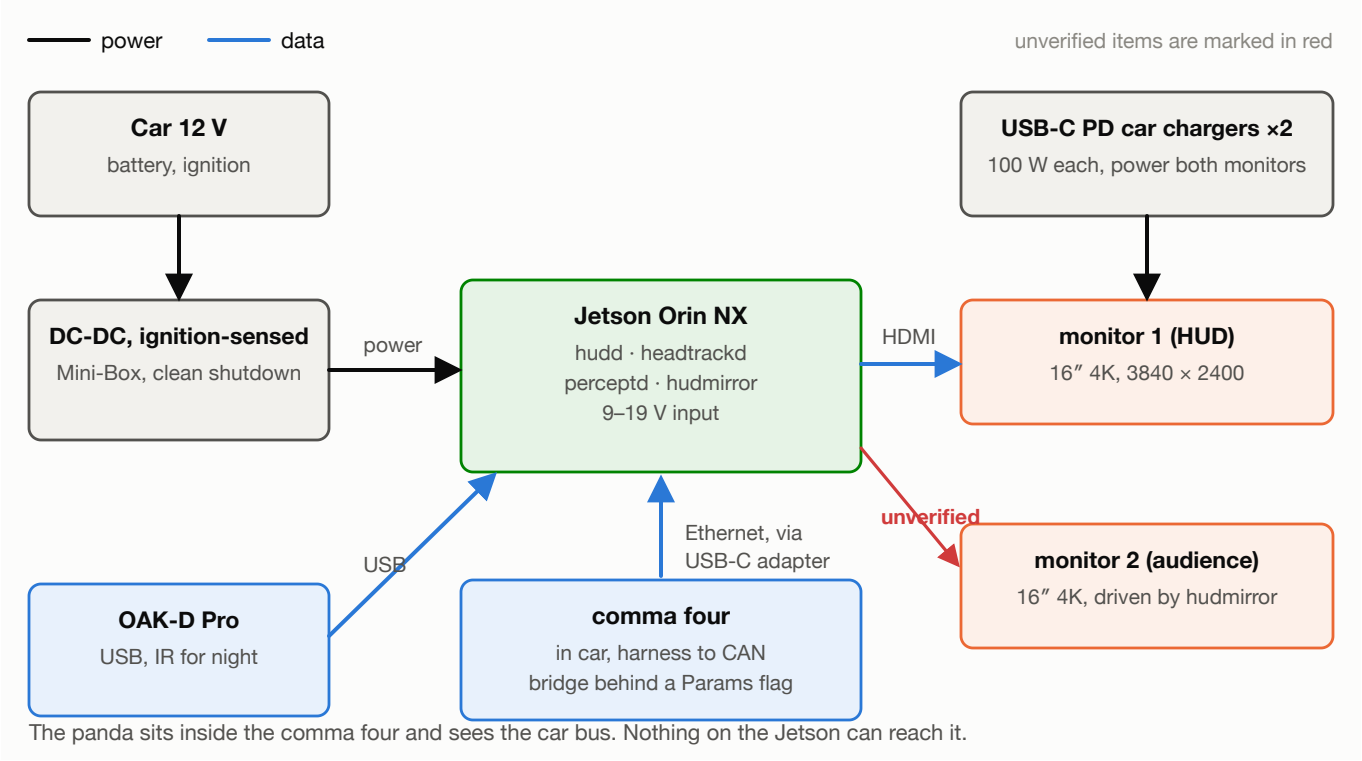


Figure 21. Power and data wiring. Power flows from the car battery through the DC-DC converter to the Jetson, and from the chargers to the monitors. Data flows by Ethernet from the comma four and by USB from the sensors. Items marked unverified are open.

One wiring question is not answered in the plan. The Jetson drives monitor 1 over HDMI, and the JetPack 5.1.3 record shows a single HDMI output in use. Monitor 2, for the audience view, needs a second output, either a second HDMI or DisplayPort on the carrier, or a USB or network path into a second display device. UNVERIFIED The carrier's display outputs must be checked before the audience view is designed.

The Ethernet path has one more check. The plan says the comma four's USB 3.1 port powers the USB-C Ethernet adapter, and whether that port runs in host mode with a suitable USB Ethernet driver has not been tested (VERIFY). If it does not, the fallback is the comma four's Wi-Fi hotspot, with the jitter cost noted in section 4.5.

8.4 Order of purchases

Revision 2 listed the orders as one block in Phase 0. Three of them depend on something else, so the plan sorts them by what each one waits on.

WHEN	WHAT TO ORDER	WHAT IT WAITS ON
Week 0–1	comma four, car harness, OAK-D Pro, USB-C PD chargers, Mini-Box, Ethernet parts, beamsplitters, Fresnel lenses, diffuser and polariser films	Car choice only for the harness; the rest does not depend on the car
Week 0–1	Request mirror quotes from optics shops	A quote does not commit the order. The spec must match the windshield, so the quote is revised after the ray trace.

WHEN	WHAT TO ORDER	WHAT IT WAITS ON
After car choice	OEM HUD-option windshield for the chosen car	Car choice, and confirmation that the windshield exists for that car
After ray trace	Custom concave mirror	The ray trace with the windshield's real curvature (section 6.6)
After input-lag test	Laser DLP projector	A measured input lag and a confirmed polarisation output. The plan says measure before buying and prefer a game mode.
Before any sun test	Thermocouple and the shutter or shade, if adopted	The sun-hazard review (section 6.7)

The rows above differ from revision 2 in two places. The mirror is quoted early and ordered late, and the projector waits for a measurement that revision 2 did not require. The first change can lengthen the Stage 2 critical path, because the mirror cannot be ordered until the ray trace and the car are done; section 9 shows where that lands. The second protects the projector budget.

9 Phases and acceptance criteria

Weeks count from the day the car and comma four are chosen. Phases overlap. The bench work in phases 1 and 2 does not wait on any purchase except the beamsplitter, so software progress is not held up by optics. Figure 22 shows the schedule, and each phase below gives its work, the reason for its position, and the test that closes it.

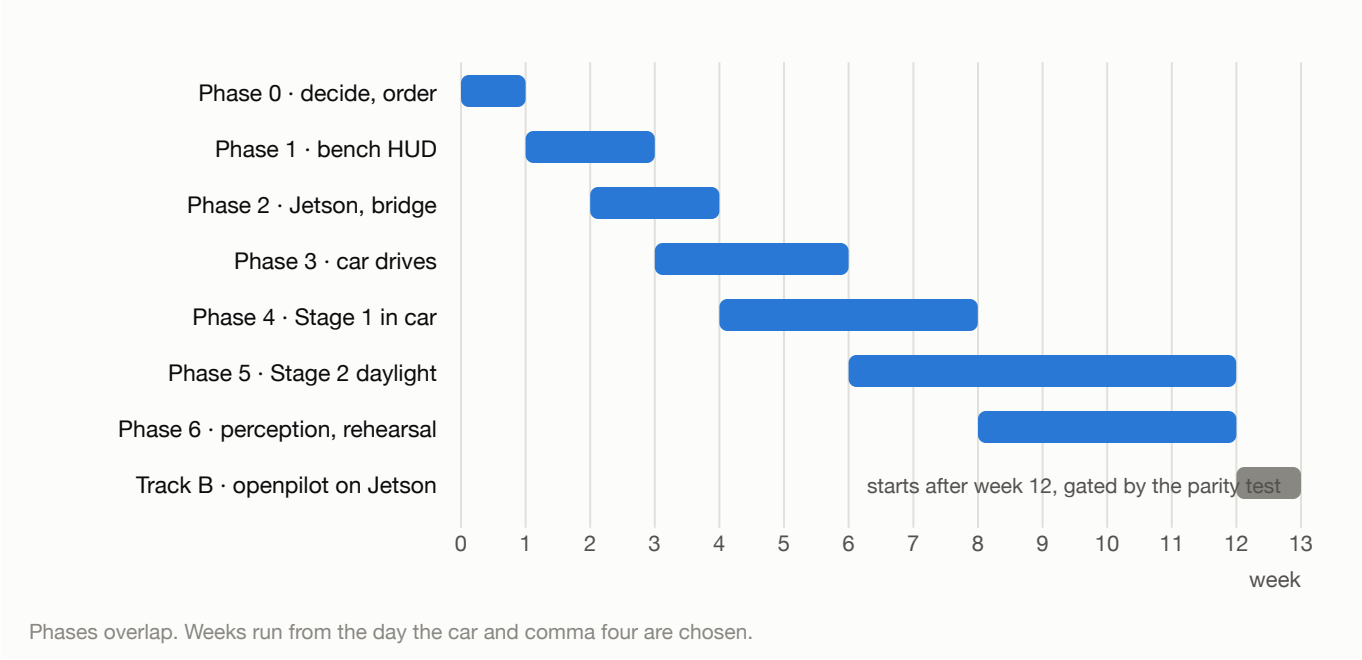


Figure 22. Phase schedule over weeks 0–13. Blue bars are phases; the grey bar is Track B, which starts only after week 12 and is gated by the parity test. Bars show the planned windows.

9.1 The critical path

Two chains determine the earliest date for the demo. The first is the software chain: the bench HUD, the Jetson, the car drive and the in-car combiner. Apart from the car, the comma four and the beamsplitters, it waits on no purchase. The second is the optics chain, and it is the one revision 2 underestimated.

The optics chain runs: car chosen, windshield identified, ray trace with that windshield, mirror order, mirror lead time, mirror arrival, Stage 2 build, daylight measurement. Revision 2 placed the mirror order in week 1 and showed Phase 5 in weeks 6–12. That would require the car to be chosen and the ray trace finished in the same first week, which the plan does not show. The mirror's lead time is also not in the plan, so the schedule cannot yet show whether week 12 holds.

> **Schedule risk, and the fallback**
> If the mirror arrives late, Phase 5 moves past week 12 and so does the demo. The plan already names Stage 1 as the demo fallback: the Fresnel magnifier and combiner, with the laser DLP projector used as the picture generator, give a working in-car demonstration without the custom mirror. Whether the Stage 1 image is good enough to be shown is open (section 11).

9.2 Gates

Each gate is a test whose result decides whether the next phase starts. A gate that fails sends the work back, and it is recorded, rather than being waived to keep the schedule.

GATE	CRITERION	DECISION IT CONTROLS
G0 · Car chosen	Car named from `docs/CARS.md`, with a factory HUD trim and a windshield confirmed to exist for it	Harness, windshield and mirror specification
G1 · Bench HUD	Path, lane lines, lead chevron and speed block render at 60 fps from replay on the Stage 0 rig	Start of Phase 2 on the Jetson
G2 · Latency	End-to-end age measured with the photodiode, under 60 ms on the monitor	Whether the projector's input lag is acceptable
G3 · Car drives	openpilot holds lane and gap on a 20-minute drive, with the Jetson showing the live path	Start of Stage 1 in the car
G4 · Stage 1 registration	Static error at 30 m ≤ 1.0° and path on the lane through curves at 25 m/s	Whether Stage 1 is shown and whether Stage 2 is pursued
G5 · Daylight	Every row of the validation table in section 10 passes at noon	Demo format: daylight or night
G6 · Rehearsal	Script runs twice in a row without an operator touching the Jetson	The demo date
G7 · Track B parity	modeld on CUDA matches the Qualcomm build within a tolerance set before the test	Whether the Jetson runs openpilot, or stays a HUD computer

9.3 Phase by phase

Week 0–1

Phase 0 · Decide and order

- Choose the car from `docs/CARS.md`, preferring one with a factory HUD trim and openpilot longitudinal control.
- Order the items in section 8.4 that do not depend on the car. Request mirror quotes, without placing the order.
- Confirm with the university who approves the vehicle modification, and where the demo may be driven.
- **Done when:** the car is named, the windshield for it is confirmed, the first orders are placed, and the demo venue is agreed.

Week 1–3

Phase 1 · Bench HUD on the mini PC

- Build the fork on mini PC 1. Replay the demo route. Write `hudd` as a new layout that reuses `ModelRenderer` with an eye-frame transform.
- Put the Stage 0 rig on the desk. Build the content from section 3, items 1 to 4.
- Define the stale-data and disengagement behaviours listed in section 3.3 before the first drive. They are needed for the rig as much as for the car.
- **Done when:** the conformal path, lane lines, lead chevron and speed block render at 60 fps from replay on the beamsplitter rig, and `tools/clip` produces a regression MP4.

Week 2–4

Phase 2 · Jetson bring-up and the bridge

- Flash JetPack 6.2 and install the fork's runtime. Run `hudd` full screen at 3840 × 2400 and 60 Hz on monitor 1.
- Run the bridge from the mini PC's replay to the Jetson. Align clocks with NTP between hosts. Instrument age in `hudd`.
- Run `headtrackd` on the OAK-D, publishing `hudEyePose`. Show the parallax correction on the bench rig.
- **Done when:** end-to-end age from replayed `modelV2` publish to photon is measured with the photodiode and is under 60 ms on the monitor. Head motion of ±75 mm moves the overlay correctly against a bench target.

Week 3–6

Phase 3 · The car drives

- Install the comma four and harness. Run the fork's branch. Complete the calibration drives. Engage ACC and lane centering.
- Add the bridge on the comma four behind a Params flag. Mount the Jetson and power supply, and make the Ethernet link live.
- **Done when:** openpilot engages and holds lane and gap on a 20-minute drive, and `hudd` on the Jetson shows the live path on monitor 1 in the passenger footwell.

Week 4–8

Phase 4 · Stage 1 combiner in the car

- Build the dash enclosure with monitor 1, the Fresnel lens and the combiner. Calibrate the eye point with the target board. Fit the pre-distortion mesh.
- Drive at night and in overcast. Show the mirror view on monitor 2.
- **Done when:** static registration error at 30 m is ≤ 1.0° (the Fresnel stage allowance), and the path stays on the lane through curves at 25 m/s.

Week 6–12

Phase 5 · Stage 2 relay mirror, daylight

- Requires the ray trace and the mirror, both after the car is chosen (section 6.6). Install the windshield. Mount the projector PGU, fold mirror and concave mirror in the printed housing. Re-calibrate.
- Measure the polarisation leaving the diffuser and fit the polariser sheet. Measure the monitor's polarisation axis and orient it, or add a retarder, before it is used as the night PGU (section 6.3).
- Measure luminance and contrast at noon. Measure the projector's dimming floor and fit a fixed neutral-density filter if it cannot reach the daylight working point (section 6.5). Build the dimming curve against the ambient sensor.
- **Done when:** every row of the validation table in section 10 passes.

Week 8–12

Phase 6 · Jetson perception, polish, rehearsal

- Run `perceptd` with a TensorRT 10 detector. Show pedestrian and sign outlines as HUD-only cues.
- Review clutter: remove or dim everything that is neither conformal nor safety-relevant.
- Rehearse the section 10 script three times on the demo course.
- **Done when:** the script runs twice in a row without an operator touching the Jetson.

After week 12

Track B · openpilot on the Jetson

- CUDA tinygrad build of the model, then output parity against replay, then USB pandad and sensors. Go/no-go after the parity test, against the tolerance set in advance (section 5.3).

9.4 What the schedule does not show

The Gantt chart shows planned windows, not measured durations. Four things are unknown and could move the dates. The custom mirror's lead time has not been quoted. The OAK-D and projector delivery times are not stated. The windshield's availability for the chosen car is unconfirmed. And the time to complete the ray trace has not been estimated. Each one should be given a date at Phase 0, and the schedule redrawn when they are known.

10 Validation and demo script

Every target in the plan is measurable, and every measurement has a method. This chapter gives the targets, the method for each, and the caveats that come with the method. The targets are the acceptance tests for the gates in section 9.

10.1 Acceptance criteria

CRITERION	TARGET	HOW IT IS MEASURED
Static registration at 30 m	≤ 0.5°	A camera at the eye point photographs a rendered marker against a surveyed target board.
Registration under head motion, ±75 mm	≤ 0.3°	Same setup, with the camera moved on a rail and the tracker on.
Motion-to-photon age, and predicted error	≤ 60 ms monitor; ≤ 100 ms projector; predicted error ≤ 0.3°	A photodiode on the PGU, triggered by a synthetic `modelV2` step. The age reported by `hudd` is cross-checked against it.
Luminance at the eye, noon	≥ 1,500 cd/m²; contrast ≥ 1.15	A spot luminance meter at the eye point, measuring a white test patch against the measured sky background.
Night luminance	≤ 50 cd/m² at minimum dim	The same meter, at the lowest setting.
Eye box	≥ 120 × 60 mm without tracker	A camera on the rail, checking that the full image is visible across the box.
Ghost image	None visible at nominal eye height	Visual, photographed.
Virtual image distance	≥ 6 m, target 7.5 m	The camera focus method, against a reference target at known distance.
Control path independence	Jetson power-cycled mid-drive; openpilot unaffected	A closed-course test. The panda's limits are unchanged by anything on the Jetson.

10.2 What each measurement can and cannot show

Most of the criteria are angles, so they are measured with a camera placed where the eye will be. A camera is not an eye. It has a fixed focus and a fixed pupil, while the driver's eye accommodates and has a different pupil. The camera therefore measures where the image falls in angle, which is what registration is, but it does not measure how the driver will perceive it. A perception test with real drivers is outside this plan.

The predicted-error target has a gap that the plan does not close. It is stated as "predicted error ≤ 0.3°", but the plan does not say how the predicted position is compared with the true one at the eye. One practical method is to place a target at a surveyed road point, move the car past it, and compare the photographed

mark with the target's position at the moment of the photodiode event. That method is proposed here and has not been agreed.

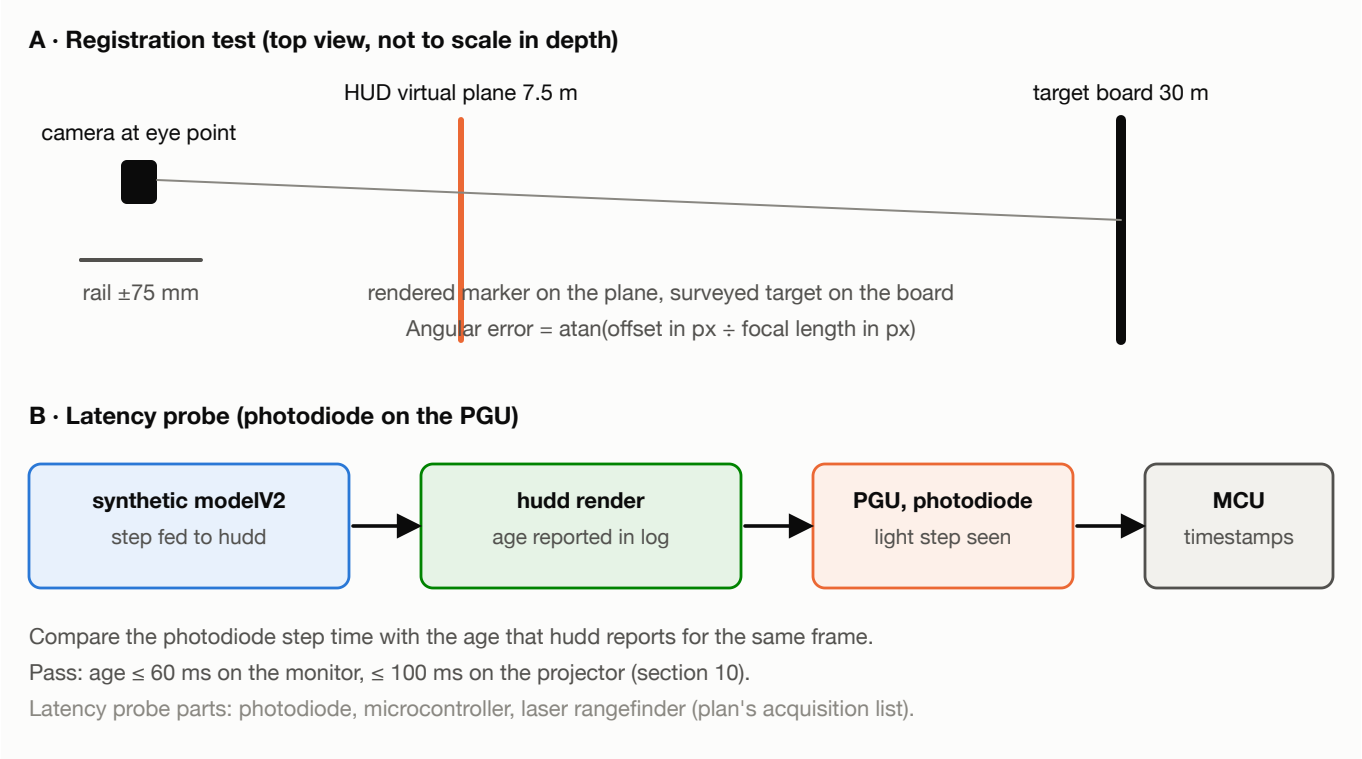


Figure 23. Two measurement setups. Top: the registration test, with the camera at the eye point on a rail and a surveyed target board at 30 m. Bottom: the latency probe, with a photodiode on the PGU timing a synthetic step. Panel A is not to scale in depth.

The luminance measurements need the meter at the eye point and the sky measured in the same direction as the HUD image. The contrast ratio is (overlay + background) ÷ background, as defined in section 6.1. Measuring the background in a different direction, or at a different time, invalidates the ratio.

10.3 The five-minute demo script

The script is the demonstration the plan builds toward. It is a sequence of six scenes, each showing one of the elements in section 3. The times are the plan's marks for the script, not timed rehearsals. A safety driver who is not the presenter drives the car on the course, as section 11 requires. The driver must not wear polarised sunglasses, and no polarising filter may sit in front of a camera at the eye point, because the s-polarised image is largely blocked by them (section 11.1).

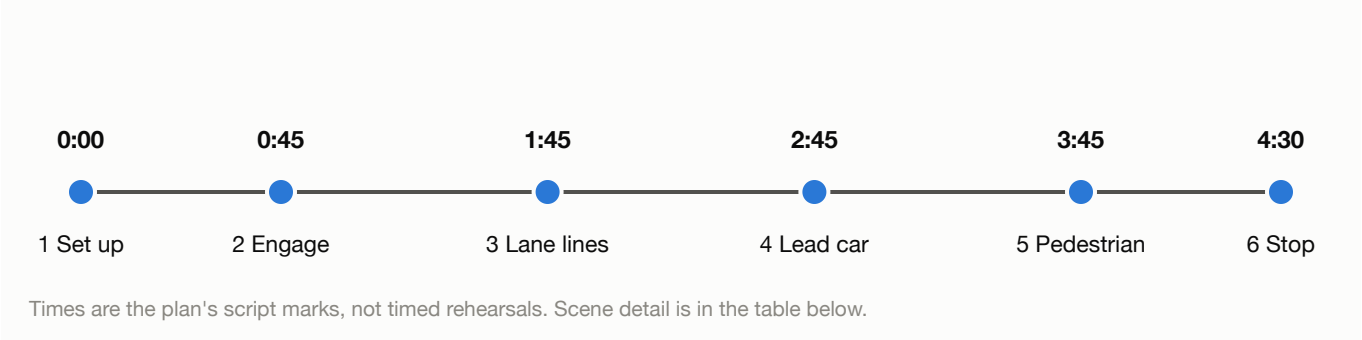


Figure 24. The five-minute demo script as a timeline: six scenes from 0:00 to 4:30. Scene detail follows in the table.

TIME	SCENE	WHAT THE AUDIENCE SEES
0:00	1 · Set up	Car stationary, HUD off. The presenter explains the two computers and the safety boundary in one breath.
0:45	2 · Engage	The driver pulls onto the course. Speed and set speed appear. openpilot engages, and the path ribbon turns green and lies on the lane.
1:45	3 · Lane lines	Lane lines fade in over the paint. The driver leans left and right, and the audience monitor shows the eye point moving with the overlay holding.
2:45	4 · Lead car	A second car is ahead. The chevron locks under its bumper with the gap, and ACC follows it through a slow-down.
3:45	5 · Pedestrian	A volunteer pedestrian at the roadside is outlined by the Jetson's detector. A deliberately cluttered mode is shown for ten seconds, then the clean mode returns.
4:30	6 · Stop	The audience monitor shows the measured registration error, age and contrast from that run.

The script ends with the measured numbers rather than a claim: at 4:30 the audience monitor shows the registration error, the age and the contrast from that run. The plan does not say what happens when a number misses its target. This edition recommends showing each target beside its number, so a miss is visible to the audience. That is a proposal, not a plan requirement.

11 Risks, legal, open questions

11.1 Risks

The first seven risks are from revision 2, with their mitigations. The rest are added in this edition, each from a gap found while checking the plan's numbers. Each added risk is tagged.

RISK	WHAT COULD HAPPEN, AND THE MITIGATION
Custom mirror lead time or quality	The mirror arrives late or does not match the ray trace. Mitigation: Stage 1 is the demo fallback, and the 70/30 combiner with the projector PGU gives daylight brightness without the windshield.
Projector input lag	Some units exceed 50 ms, which pushes the motion-to-photon age over target. Mitigation: measure before buying, prefer a model with a game mode. Prediction covers part of the lag (section 4.6).
Head tracker at night	Tracking quality at night is not yet characterised. Mitigation: the OAK-D Pro's IR illuminator is the reason for choosing it. MediaPipe degrades past 45° of yaw, which is acceptable for a driver looking ahead.
Sunlight focused onto the PGU	The concave mirror concentrates sunlight near the diffuser (section 6.7). Mitigation in the plan: cold mirror and IR-cut glass, and the housing aperture shaded by the dash lip. The "cold mirror" term is wrong for this purpose, and filters do not remove the focus. Added: thermocouple test before any sun exposure, and a shutter proposed for review.
Jetson state unknown	The facts are from 2026-10-07: JetPack 5.1.3, 15 W mode, no camera, and a 3840 × 2400 panel on HDMI. Mitigation: verify before flashing.
Bridge port collisions	Ports are hashed from service names, with a known FIXME. Mitigation: the whitelist is short, so collisions are checked once at setup.
Attention capture	Superimposed symbology draws attention away from the scene; aviation research on HUDs is the basis. Mitigation: keep everything conformal and near the lead vehicle, with no full-windshield content.
Relay geometry depends on the windshield ADDED	The single-mirror calculation cannot place the image correctly without the windshield in the model. Mitigation: a ray trace with the chosen car's windshield before the mirror is ordered (section 6.6).
Dual-monitor output unverified ADDED	The Jetson may not have a second display output for the audience monitor. Mitigation: check the carrier's outputs before designing the audience view (section 8.3).
Prediction sign or rotation wrong ADDED	A sign error would move the path in the wrong direction, by an amount comparable to the uncompensated error, about 3° at 10 m. Mitigation: confirm the sign on a replayed route before any car drive (section 4.6).
Parity tolerance undefined ADDED	Track B's go/no-go could be argued either way. Mitigation: set a numeric tolerance before the test (section 5.3).
Stale or missing data ADDED	The HUD could draw the last known path after its source has stopped. Mitigation: define the stale-data cutoff and the disengagement behaviour before Phase 1 (section 3.3).

RISK	WHAT COULD HAPPEN, AND THE MITIGATION
Ghost image off the nominal eye height ADDED	The wedge removes the ghost only at the design eye point, so the driver may see a double image when the head moves. Mitigation: the eye-box test and the ghost photograph at several heights.
Polarised sunglasses ADDED	The design relies on an s-polarised image, and polarised sunglasses are made to block exactly that polarisation, so a driver wearing them sees a dim or absent HUD. Production HUDs that tolerate sunglasses use p-polarised glass with a special coating (AGC, in the sources), which is out of this plan's scope. Mitigation: no polarised eyewear for the driver or in front of the eye-point camera during the demo, stated in the script (section 10.3).
Monitor polarisation orientation ADDED	An LCD monitor emits linearly polarised light. Mounted in the wrong orientation, its windshield reflection falls from 23.6% to 1.2%. Mitigation: measure the panel's axis and orient it, or add a half-wave retarder, before it is used as the night PGU (section 6.3).
Polariser loss ADDED	The diffuser depolarises the projector's light, so a polariser sheet is needed and passes about 42% of it. Daylight headroom falls from 23× to 10×, which still passes. Mitigation: carried in the budget; measure the diffuser's output before assuming the loss can be avoided (section 6.5).
Projector dimming floor ADDED	Daylight needs about 10% of full output, and some laser projectors cannot dim that far by source control. Mitigation: measure the floor; a fixed neutral-density filter sets the daylight working point if needed (section 6.5).

11.2 Legal CONFIRM LOCALLY

No federal standard governs head-up displays directly. NHTSA has treated factory head-up display areas as minor exceptions to the defrost and wiper zones. State law governs objects on or near the windshield. California Vehicle Code 26708 bars anything that obstructs the driver's clear view, with small exempt zones in the lower corners and top centre. Texas and Florida have similar obstruction statutes. These statements are from revision 2 and have not been re-checked for this edition.

The practical question is which configuration is least exposed. A dash-top housing that projects into the glass without attaching anything to it is the least exposed configuration. A free-standing combiner above the dash is more exposed, because it stands in the driver's view. The Stage 2 housing is of the dash-top kind, which the plan treats as least exposed. The Stage 1 combiner stands above the dash and is the more exposed of the two. Both still need the local check, and the Stage 1 combiner is the riskier stage for public use even though it is the simpler one.

For the demo, the plan's conditions are a closed course or private lot, a safety driver who is not the presenter, and sign-off from the university's risk office before any public road use. Those conditions are the plan's, not legal advice, and the university's rules govern.

11.3 Open questions

1. **Which car.** This gates the harness, the windshield and the quality of longitudinal control, and it also fixes the windshield curvature that the mirror design depends on.
2. **Whether the classic J401 carrier accepts a GMSL deserialiser board,** or whether `perceptd` uses a USB camera instead.

3. **Whether the demo venue allows** a second vehicle as the lead car and a volunteer pedestrian.
4. **Whether the Stage 1 image quality is good enough to be shown,** or is for development only.
5. **How the Jetson drives two monitors,** one for the HUD and one for the audience (section 8.3).
6. **What the ray trace gives** for f, u and the mirror shape, with the chosen windshield (section 6.6).
7. **What tolerance defines "matching"** for Track B's parity test (section 5.3).
8. **What the HUD does on stale data and on disengagement** (section 3.3).
9. **How the sun test is run,** and whether the shutter or shade is adopted (section 6.7).
10. **Which term the optics vendor uses** for the infrared-blocking component in front of the diffuser.
11. **How far the chosen projector can dim,** and whether a fixed neutral-density filter is needed for the daylight working point (section 6.5).
12. **What polarisation leaves the diffuser,** and whether a wire-grid polariser recovers enough light over an iodine film to be worth its cost (section 6.3).

A Derivations

Each formula used in the plan, with the worked numbers. The values are the ones printed in the text and computed in the build script, so a reader can check any figure by hand.

A.1 Fresnel reflectance

For light from air ($n_1 = 1$) reaching glass ($n_2 = 1.52$) at incidence angle θ_i, with the transmitted angle θ_t from Snell's law, the reflected fractions are:

$$\sin\theta_t = \sin\theta_i \div n_2$$
$$R_s = [(\cos\theta_i - n_2\cos\theta_t) \div (\cos\theta_i + n_2\cos\theta_t)]^2$$
$$R_p = [(n_2\cos\theta_i - \cos\theta_t) \div (n_2\cos\theta_i + \cos\theta_t)]^2$$

At $\theta_i = 65°$: $\sin\theta_t = 0.906 \div 1.52 = 0.596$, so $\theta_t = 36.6°$. Then $R_s = 23.6\%$ and $R_p = 1.2\%$. R_p is zero at the Brewster angle, arctan(1.52) = 56.7°. These are per surface; a laminated windshield has two.

A.2 Mirror equation, Stage 2

For a concave mirror of focal length f, an object at distance u in front of the mirror forms an image at distance v, with $1/u + 1/v = 1/f$. Magnification is $M = -v/u$. When the object is inside the focal length, v is negative and the image is virtual, on the object's side, and magnified.

Virtual image at 7.5 m, f = 330 mm:
1/u = 1/f + 1/7500 = 0.0030303 + 0.0001333 so u = 316.1 mm
M = 7500 ÷ 316.1 = 23.7
image width = 60 mm × 23.7 = 1.42 m so field 2·atan(0.711 ÷ 7.5) = 10.8°
image height = 30 mm × 23.7 = 0.71 m so field 2·atan(0.356 ÷ 7.5) = 5.4°

Revision 2's u = 346 mm, f = 330 mm:
1/v = 1/f − 1/u = 0.0030303 − 0.0028902 = 0.0001401 so v = +7.14 m (real)
|M| = 7140 ÷ 346 = 20.6

The second case is the one revision 2 had. Its positive v means a real image, 7.1 m from the mirror, which cannot be seen as a virtual image at 7.5 m. The single-mirror model also omits the windshield, which is covered in section 6.6.

A.3 Lens equation, Stage 1

A thin converging lens of focal length f, with an object at distance u on the object side, follows the same equation. With f = 300 mm and u = 280 mm:

1/v = 1/300 − 1/280 = −0.000238 so v = −4200 mm (virtual, object side)
M = 4200 ÷ 280 = 15
image = 80 mm × 15 = 1.2 m wide, 40 mm × 15 = 0.6 m high
field = 2·atan(0.6 ÷ 4.2) = 16.3° wide, 2·atan(0.3 ÷ 4.2) = 8.2° high
pixels per degree: (80 mm ÷ 0.0917 mm per px) ÷ 16.3° = 872 ÷ 16.3 ≈ 54

A Fresnel lens has the same equation as the thin lens it approximates. The grooves change the aberrations, which is why the plan expects ring artefacts.

A.4 Window size

The width of a window of field θ at distance D is 2D·tan(θ/2). At 7.5 m, the 10° window is 2 × 7.5 × tan 5° = 1.31 m wide. The 4° height is 2 × 7.5 × tan 2° = 0.52 m, and the 5° height is 0.65 m. The corrected 10.8° × 5.4° window is 1.42 m × 0.71 m.

A.5 Luminance from flux, and the budget chain

For a Lambertian emitter of area A and luminance L, the total flux into the hemisphere is πLA. Inverting it, a transmissive diffuser that passes a fraction η of flux Φ has luminance $L = \Phi\eta \div (\pi A)$. For Φ = 2000 lm, η = 0.5, and A = 0.060 × 0.030 = 0.0018 m²:

```
L = 2000 × 0.5 ÷ (π × 0.0018) = 176,839 cd/m²
```

Each optical element then multiplies the luminance by its reflectance or transmittance. An ideal imaging mirror changes the image size but not its luminance, so the chain is a product of fractions:

```
Combiner path:          L_eye = L_source × 0.30                  = 53,052
Windshield path:        L_eye = L_source × 0.24                  = 42,441
Windshield + 2 mirrors  L_eye = L_source × 0.90 × 0.90 × 0.24    = 34,377
With polariser sheet    L_eye = L_source × 0.90 × 0.90 × 0.24 × 0.42 = 14,439
```

The 0.90 mirror figure and the 0.42 polariser transmission are assumptions. The polariser is needed because a rear-projection diffuser depolarises the light that passes through it (section 6.3). The windshield factor is the s-polarised R_s from A.1 for one surface, rounded to 0.24. A laminated windshield reflects from both of its air-glass surfaces: the first reflects R_s, and the second returns $(1 - R_s)^2 R_s$ after two passes through the first. With the wedge overlapping the two images at the design eye point, the combined reflection is about 37%, so 0.24 is the conservative figure.

A.6 Contrast ratio and the noon target

The plan defines contrast as (overlay + background) ÷ background. For a background of 10,000 cd/m² and a required ratio of 1.15, the overlay must supply 0.15 × 10,000 = 1,500 cd/m² at the eye.

A.7 Dimming range

```
day target ÷ night ceiling      = 1,500 ÷ 50    = 30:1
full output ÷ night ceiling     = 42,441 ÷ 50   = 849:1
daylight fraction, bare windshield path    = 1,500 ÷ 42,441 = 3.5%
daylight fraction, mirrors and polariser   = 1,500 ÷ 14,439 = 10%
```

A.8 Parallax from head shift

A head shift of Δ moves the line of sight to a point at distance d by an angle atan(Δ ÷ d), if the image is not corrected. For Δ = 0.150 m: at 7.5 m, atan(0.02) = 1.15°; at 10 m, atan(0.015) = 0.86°; at 30 m, atan(0.005) = 0.29°.

A.9 Motion compensation

A camera at height h = 1.22 m sights a road point at horizontal distance d. The depression angle is θ = atan(h ÷ d). At 30 m/s, the car travels 3.0 m in 100 ms, so the point's distance drops from 10 m to 7.0 m:

Uncompensated: θ(10 m) = atan(1.22 ÷ 10) = 6.96°
θ(7.0 m) = atan(1.22 ÷ 7.0) = 9.89°
change = 2.9°

Residual after prediction, for unmodelled acceleration a = 1 m/s² over t = 0.1 s:
$\Delta x = \frac{1}{2} a t^2 = 5.0$ mm
$\Delta\theta \approx h \div (d^2 + h^2) \times \Delta x = 1.22 \div 101.5 \times 0.005 = 6.0 \times 10^{-5}$ rad = 0.003°

The residual is small because prediction removes the large term, the car's own travel. What remains is what the model did not foresee.

A.10 Latency sums

The four timed stages sum to a minimum of 50 + 1 + 8 + 8 = 67 ms and a maximum of 80 + 3 + 16 + 40 = 139 ms. At 30 m/s these are 2.0 m and 4.2 m of travel, which the plan rounds to 2 to 4 m. The head tracker is excluded because it affects parallax, not the road-frame path.

A.11 Where a road point lands on the virtual plane

For a flat road and an eye at height h, the line of sight to a point at distance d crosses a plane at distance D at height h(1 − D ÷ d) above the road. With h = 1.22 m and D = 7.5 m: 1.22 × (1 − 7.5 ÷ 10) = 0.31 m for d = 10 m, and 1.22 × (1 − 7.5 ÷ 30) = 0.92 m for d = 30 m.

A.12 Sun focus estimate

The sun subtends 0.53° (9.25 mrad). Its focused image at the focal plane is about f × 9.25 mrad = 330 × 0.00925 = 3.1 mm across. For the ray-trace estimate, parallel on-axis rays strike a spherical mirror of radius 660 mm over a 300 × 200 mm aperture. Each reflected ray is intersected with the PGU plane 316 mm from the vertex. The radius that contains 90% of the rays is 3.2 mm at that plane. The ideal, lossless concentration is the aperture area over the area of that spot:

$C = (300 \times 200\ \text{mm}^2) \div (\pi \times r90^2) \approx 1{,}869$ (r90 = 3.2 mm, unrounded in the calculation)

The trace gives a larger spot, 4.1 mm, at the paraxial focal plane, because marginal rays focus nearer the mirror (spherical aberration). The PGU plane is therefore close to the smallest blur, which is why it is the worst location for the diffuser. The estimate ignores the windshield and the sun's disk, so it is an upper bound.

```
collected power = 0.300 m × 0.200 m × 1,000 W/m² × 70% = 42 W
spot diameter ≈ 2 × r90 = 6 mm at the PGU plane
```

The 1,000 W/m² is direct sunlight on a clear day, and the 70% is an assumed windshield transmission for sunlight that covers both surfaces and the interlayer. The aperture is taken as fully lit, which is the worst case; the dash lip and the housing will shade part of it at most sun positions.

A.13 Acquisition totals

The sum of the low ends of the cost ranges in section 8.2 is $8,552, and the sum of the high ends is $16,192. The total excludes the deferred ZED-F9P, and it includes single prices and the mirror quote as listed.

B Glossary

TERM	MEANING IN THIS PLAN
ACC	Adaptive cruise control. Holds a set speed and a gap to the car ahead.
Accommodation	The eye's change of focus with distance.
Age (of a message)	Current time minus the message's timestamp, here `modelV2.timestampEof`.
Beamsplitter	Partly reflecting, partly transmitting glass. A 70/30 part transmits 70% and, in the teleprompter convention, reflects 30%.
Brewster angle	The incidence angle at which the p-polarised reflection of a surface is zero.
Bridge	The openpilot program that converts between local msgq and network ZMQ.
CAN bus	The car's internal network over which the control units exchange messages.
Combiner	A partly reflecting glass that merges the image from a display with the view through it.
Conformal	Graphics that must sit on the thing they describe, such as the drive path on the road.
Contrast ratio	In this plan, (overlay + background) ÷ background. The noon target uses 1.15.
Desync	Two messages describing different moments. The age compensation exists to prevent it.
DC-DC converter	Here, the ignition-sensed supply that powers the Jetson and lets it shut down cleanly.
Eye box	The region the eye can move through while still seeing the whole image.
Field of view	The angle the image covers as seen from the eye.
Fresnel equations	The formulas for reflected and transmitted fractions at a boundary between two media.
Fresnel lens	A thin lens whose power comes from concentric grooves. Used here as a magnifier in Stage 1.
GMSL	Gigabit Multimedia Serial Link, a camera interface used by automotive cameras.
Keystone	A trapezoidal distortion of an image from an off-axis mirror. Corrected in software by hudd.
Lambertian	A surface whose luminance is the same in every direction.
Lane centering	openpilot's steering that keeps the car within its lane.
Luminance (cd/m²)	Brightness of a surface as seen by the eye. Also called a nit.
modelV2	The driving model's output message: path, lane lines, lead predictions. Published at 20 Hz.
msgq	openpilot's local inter-process message queue.
Motion-to-photon latency	Delay from the moment the scene changes to the moment the image shows it.

TERM	MEANING IN THIS PLAN
Nominal eye point	The design eye position, from which the calibration is measured.
Panda	The board in the comma four that connects the computer to the car's CAN bus and enforces the safety model in firmware.
Parallax	The apparent shift of an object when the viewpoint moves.
Params	openpilot's persistent key-value store for settings and calibration.
PGU	Picture generation unit: the device that makes the image, here a laser DLP projector on a diffuser or a monitor.
Pre-distortion	A warp applied to the image so that the optics' own distortion cancels it.
PVB, wedge PVB	Polyvinyl butyral, the plastic interlayer in laminated glass. A wedge-shaped version removes the ghost image at the design eye point.
Registration	How accurately a graphic lines up with the thing it marks.
Relay mirror	A mirror that re-images the PGU's picture into the windshield. Stage 2.
s- and p-polarisation	The two perpendicular polarisations of light. The windshield reflects s much more strongly.
Safety model	The per-car set of limits that the panda's firmware enforces on every control frame.
SubMaster	A message subscriber that keeps the latest message from each service it reads.
Track A, Track B	Track A: the comma four drives. Track B: openpilot runs on the Jetson, a research track.
Virtual image	An image that the eye perceives at a distance, though no light originates there.
ZMQ	ZeroMQ, the network messaging library the bridge uses.

C Repository pointers

The files the HUD work touches. Line references were checked against the checkout on 2026-10-08 where marked; the rest are as written in revision 2.

FILE	WHY, AND WHAT WAS CHECKED
`openpilot/cereal/messaging/bridge.cc`, `bridge_zmq.cc`, `msgq_to_zmq.cc`	Cross-host transport and port hashing. Whitelist match at `bridge.cc` line 13: **CHECKED**.
`openpilot/cereal/services.py`, `log.capnp`	Service rates, all checked. Add `hudEyePose` and `hudDetections`.
`openpilot/selfdrive/ui/onroad/model_renderer.py`	Point lists, ground offset, clipping constants, `set_transform`.
`openpilot/selfdrive/ui/onroad/augmented_road_view.py`	The camera-frame matrix hudd replaces.
`openpilot/selfdrive/ui/onroad/hud_renderer.py`	Set-speed and speed fields.
`openpilot/selfdrive/ui/ui_state.py`	Subscriptions, the `started` logic and local Params.
`openpilot/system/ui/lib/application.py`	Window, FPS, vsync and scale. Add monitor selection and full screen.
`openpilot/common/transformations/camera.py`	Frame conventions and `DEVICE_CAMERAS`.
`openpilot/system/manager/process_config.py`	Add hudd, headtrackd, perceptd and hudmirror. The bridge is gated by `notCar` at line 118: **CHECKED**. The Track B rows are **CHECKED**: `timed` line 83 and `sensord` line 88 are disabled on PC, `qcomgpsd` line 101 and `ubloxd` / `pigeond` lines 105–106 are comma-hardware only, and `hardwared` at line 111 runs everywhere.
`openpilot/selfdrive/modeld/SConscript`, `helpers.py`	Track B only. The CUDA device branch sits at lines 27–31 of the SConscript: **CHECKED** (revision 2 said 28–32).
`openpilot/selfdrive/pandad/panda.cc`	Track B only. The SPI handle is constructed at line 16: **CHECKED** (revision 2 said 15).
`openpilot/tools/replay`, `tools/clip`, `tools/sim`	Bench development and regression.

D Sources

The sources cited in revision 2. They were not re-read for this edition, except where a claim is marked as checked in Appendix C. Prices and product claims should be checked again before ordering.

- Seeed, reComputer J4012 JetPack flashing guides (JetPack 6.0, 6.1 and 6.2 listed; 9–19 V input; MAXN SUPER warning): wiki.seeedstudio.com/reComputer_J4012_Flash_Jetpack and wiki.seeedstudio.com/flash/jetpack_to_selected_product, read 2026-10-08.
- NVIDIA, JetPack 6.2 Super Mode blog, 2025-01-16. JetPack 6.2 components CUDA 12.6 and TensorRT 10.3, via PyTorch and RidgeRun documentation.
- comma.ai shop, comma four and chestnut pages, read 2026-10-08.
- Arducam GMSL quick-start for Jetson (JetPack 6.2 / L4T 36.4.3), docs.arducam.com.
- TI DLP application notes DLPA057, DLPA073 and DLPA117 on AR HUD virtual image distance and field of view.
- Mercedes supplier portal and VW newsroom on S-Class and ID.4 AR HUDs (10 m); Huawei IAA 2021 release (7.5 m, 13° × 5°); Corning and Hyundai Mobis on Ioniq 5 (7.5 m); Panasonic CES 2022 AR HUD 2.0 with eye tracking.
- Analog Devices design note on HUD daylight contrast (ratio 1.15–1.5 against up to 10,000 cd/m² ambient); Karl Guttag, CES 2018 HUD notes on source luminance.
- SAE 2016-01-1402 and Eastman Saflex and Kuraray Trosifol literature on wedge PVB; AGC on p-polarised HUD glass.
- NASA Ames (Foyle; Dowell) on attention capture with superimposed HUD symbology; AIST simulator study on lead-vehicle-centred cues; Frontiers 2019 AR DriveSim study.
- arXiv 2607.23901 and Jerald 2010 on perceptual latency thresholds; Holloway 1997 on latency as the dominant registration error.
- California Vehicle Code 26708; Texas Transportation Code 547.613; Florida Statutes 316.2952; NHTSA interpretation aiam4336.
- Mini-Box DCDC-USB-200 product page; SparkFun GPS-RTK2; Waveshare OAK-D Pro listing.

This document supersedes the earlier plan and the TODOs in `.github/README.md`.